The Kronos Fusion Compact Tokamak

A High-Field, Aneutronic Fusion Reactor Architecture

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Detailed Technical Appendix Available Upon Request
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Abstract

Kronos S.M.A.R.T. (Superconducting, Minimum-Aspect-Ratio Tokamak) establishes a new technical paradigm for aneutronic fusion by combining ultra-high magnetic fields, compact spherical geometry, and a materials-first design philosophy. The architecture is constrained by the brittle nature of ReBCO high-temperature superconductors, which irreversibly degrade above 0.4% strain. This limit defines every structural choice: high-modulus carbon fibers ($E \geq 400$ GPa), MP35N substrates ($\sigma_\gamma \approx 1.7$ GPa), graphene-reinforced aluminum conductors, and refractory ceramic coatings are deployed to transform mechanical fragility into engineered stability. Supporting structures—including preloaded carbon-fiber cocoons, ceramic bucking cylinders, and cryogenic pistons—maintain equilibrium under multi-gigapascal Lorentz stresses.

The performance of the system scales with the fundamental tokamak relation:

$$P_{\text{fusion}} \propto \beta^2 B^4 V,$$

where plasma β is the ratio of plasma to magnetic pressure, BBB the toroidal field, and VVV the plasma volume. By employing a quasi-spherical aspect ratio ($A \approx 2.0$), high elongation ($\kappa > 2.5$), and negative triangularity ($\delta < 0$), S.M.A.R.T. maximizes both β and volume while sustaining ultra-high fields ($B_0 = 13.5$ T at $R_0 = 2.0$ m; B_peak > 30 T at r = 0.9 m). This yields a fusion power density several orders of magnitude higher than D–T tokamaks constrained to ≤11 T, without exceeding ReBCO's strain threshold.

The curated aneutronic fuel cycle initiates with $D + {}^3He \rightarrow {}^4He + p$ ($Q \approx 18.3$ MeV), transitions to $p + {}^6Li \rightarrow {}^4He + {}^3He$ ($Q \approx 4.0$ MeV), and supports advanced steady-state modes such as ${}^3He + {}^3He \rightarrow {}^4He + 2p$ and $p + {}^{11}B \rightarrow 3{}^4He$ ($Q \approx 8.7$ MeV). These reactions yield predominantly charged products, enabling direct energy conversion (DEC) via electrostatic collectors, MHD channels, thermionic emitters, and photon-absorbing GaAs/perovskite photovoltaics. Over 95% of the fusion energy is recoverable in charged particles and photons, avoiding the 14.1 MeV neutron flux of D–T fusion that necessitates ≥1 m shielding and frequent component replacement.

Electrical efficiency is defined as:

$$\eta_{\text{net}} = \frac{P_{\text{electrical}}}{P_{\text{fusion}}},$$

with conventional Rankine systems limited by Carnot efficiency (~35%). By integrating multi-path DEC and hybrid Brayton/Rankine loops, S.M.A.R.T. is engineered to exceed ηnet>0.60\eta_\text{net} > 0.60ηnet>0.60, cutting waste heat by more than a factor of three compared to neutron-based systems of equivalent electrical output.

Ignition and sustainment employ a redundant suite: high-efficiency neutral beam injection (300–500 keV fast atoms), adiabatic compression ($\Phi = \int B \cdot dA$ conservation), and dual-stage Ohmic Heating (OH-1: fast breakdown;

OH-2: long-pulse sustainment at ~3× efficiency). Non-insulated HTS windings further mitigate quench events by enabling passive current redistribution without destructive terminal voltages.

This design directly addresses the energy challenge of the coming decade. Global data infrastructure alone is projected to exceed 1,000 TWh annually by 2030, with hyperscale AI clusters requiring multi-GW, low-latency baseloads. Such demands cannot be met by intermittent renewables or neutron-intensive fusion. The S.M.A.R.T. reactor, compact, aneutronic, and neutron-immune by design, enables deployment in urban grids, hyperscale data centers, defense platforms, and deep-space missions.

By merging scaling-law-optimized plasma physics, materials-constrained engineering, and solid-state direct conversion, S.M.A.R.T. redefines fusion architecture: not a marginal improvement on legacy D–T systems, but a deliberate reinvention of fusion as a compact, efficient, and sustainable energy platform. It is fusion designed not around compromise, but around the equations that govern performance, survivability, and efficiency

KRONOS
FUSION ENERGY

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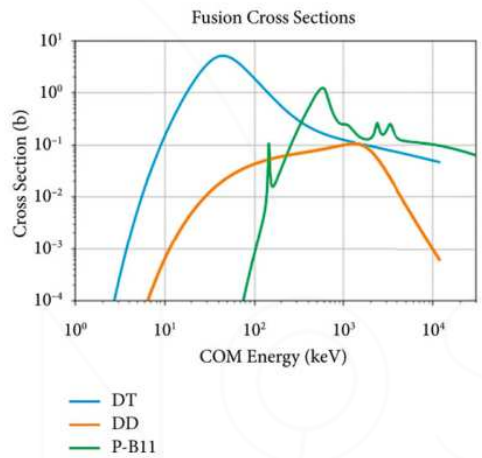
Introduction

Fusion energy offers the prospect of a virtually limitless and clean power source, but mainstream efforts have centered on the deuterium–tritium (D–T) reaction despite its drawbacks. The D–T fuel cycle has a relatively low ignition temperature (~10 keV) which eases plasma initiation, but it releases the majority of energy in 14.1 MeV neutrons. These neutrons impose severe engineering penalties: they damage reactor structures, activate materials into radioactive waste, and require thick (~1 m or more) shielding and breeding blankets to absorb energy and regenerate tritium. In a D–T tokamak, the need to protect superconducting magnets from neutron heating (and to keep them cryogenic) forces a large device size and complex cooling, and even then component lifetimes are limited by neutron embrittlement. These factors increase the cost, complexity, and downtime of D–T based reactors.

Aneutronic fusion reactions, in contrast, produce little to no neutron radiation. Reactions such as deuterium–helium-3 (D + ^{3}He), proton–lithium-6 (p + ^{6}Li), and proton–boron-11 (p + ^{11}B) release energy predominantly in charged particles (alpha particles and protons) rather than neutrons. If these reactions can be harnessed, the benefits are transformative: minimal radioactive waste and structural damage, no need for tritium breeding, and the possibility of converting fusion energy directly into electricity without a thermodynamic steam cycle. The challenge is that aneutronic fuels require much higher plasma temperatures (hundreds of keV range) and superior confinement to achieve net energy gain, because their fusion cross-sections peak at higher energies and are generally lower than D–T. Until recently, pursuing aneutronic fusion was considered impractical. However, advances in high-field magnet technology, plasma shaping, and materials science are opening new avenues to revisit aneutronic approaches.



Charts 1: Fusion reaction cross sections versus center-of-mass (COM) energy. The plot compares D–T (blue), D–D (orange), and p–^{11}B (green) fuels. D–T achieves the highest cross section at relatively low energies, while D–D has a lower peak and requires higher energies. The aneutronic p–^{11}B reaction peaks in the keV–MeV range, offering clean charged-particle outputs despite lower cross sections.

This paper presents the design of a high-field spherical tokamak, called S.M.A.R.T., explicitly optimized for aneutronic fusion. The S.M.A.R.T. concept is built around three core imperatives: (1) eliminating neutron production at the source by using aneutronic fuel cycles, (2) leveraging high-temperature superconductors (HTS) to create unprecedented magnetic field strength in a compact geometry, and (3) integrating direct energy conversion (DEC) technologies into the reactor structure to efficiently convert fusion kinetic energy into electricity. By starting from these premises, the design diverges significantly from conventional fusion reactors. S.M.A.R.T. uses a tight-aspect-ratio (approx. 2:1) spherical tokamak configuration to maximize plasma pressure (β) and magnetic efficiency, enabling a high fusion power density in a small volume. The use of REBCO (Rare-Earth Barium Copper Oxide) HTS tapes for the toroidal field (TF) coils allows operation at toroidal magnetic fields above 30 Tesla – far beyond the limits of traditional superconductors – which strongly enhances confinement and fusion reaction rates (fusion power scales roughly as B^4 in tokamaks). At the same time, the brittleness of ceramic HTS tapes under strain is addressed by a "materials-first" engineering strategy: every aspect of the magnet and structure is designed to keep tensile strain in the superconductor below 0.4%, the threshold beyond which ReBCO irreversibly loses critical current. To achieve this, S.M.A.R.T. employs a suite of novel structural supports and high-modulus materials to contain electromagnetic forces.

Another distinguishing feature of S.M.A.R.T. is its approach to energy conversion. Instead of treating electricity generation as an afterthought (as in conventional designs that simply produce heat for a turbine), this architecture treats every charged fusion product and photon as an opportunity for direct electricity generation. The reactor's inner walls and exhaust systems are lined with direct energy converters: magnetohydrodynamic channels, electrostatic charge collectors, thermionic emitters, and photovoltaic layers, all working in concert to capture the kinetic and radiative energy of the fusion plasma and convert it to electric power. This not only boosts overall efficiency but also reduces the thermal load on the reactor, since less of the fusion energy ends up as waste heat.

Figure 1: Key differentiators driving Kronos Fusion Energy's innovation roadmap—featuring aneutronic fusion, direct energy conversion, modular reactor design, rapid digital prototyping, scalable power output, and a world-class multidisciplinary team.

In the following sections, we detail the S.M.A.R.T. design and methodology, present anticipated performance metrics based on analytical modeling of the plasma and engineering structures, and discuss the trade-offs and implications of this approach. We compare the S.M.A.R.T. architecture with contemporary high-field D–T tokamak projects such as MIT's SPARC and ARC, highlighting how our design addresses the challenges of high-field magnets and aneutronic fuel in a different way. Through this, we aim to demonstrate that an aneutronic, high-field tokamak is a technically plausible path toward practical fusion energy, warranting deeper investigation by the fusion research community.

Methodology

Core Design Principles: The S.M.A.R.T. reactor is conceived from first principles to meet the demanding criteria of aneutronic fusion. At its foundation is the choice of fuel. The reactor's fuel cycle begins with a deuterium–helium-3 mixture for prompt ignition and then predominantly runs on proton–lithium-6 reactions, with the flexibility to utilize proton–boron-11 in steady-state operation. In these reactions ($D + {}^{3}He \rightarrow {}^{4}He + p$, $p + {}^{6}Li \rightarrow {}^{4}He + {}^{3}He$, and $p + {}^{11}B \rightarrow 3\ {}^{4}He$), nearly all the fusion energy is released in charged particles (helium nuclei and protons) rather than neutrons. By "designing out" neutron production, we eliminate the root cause of most fusion reactor material problems.

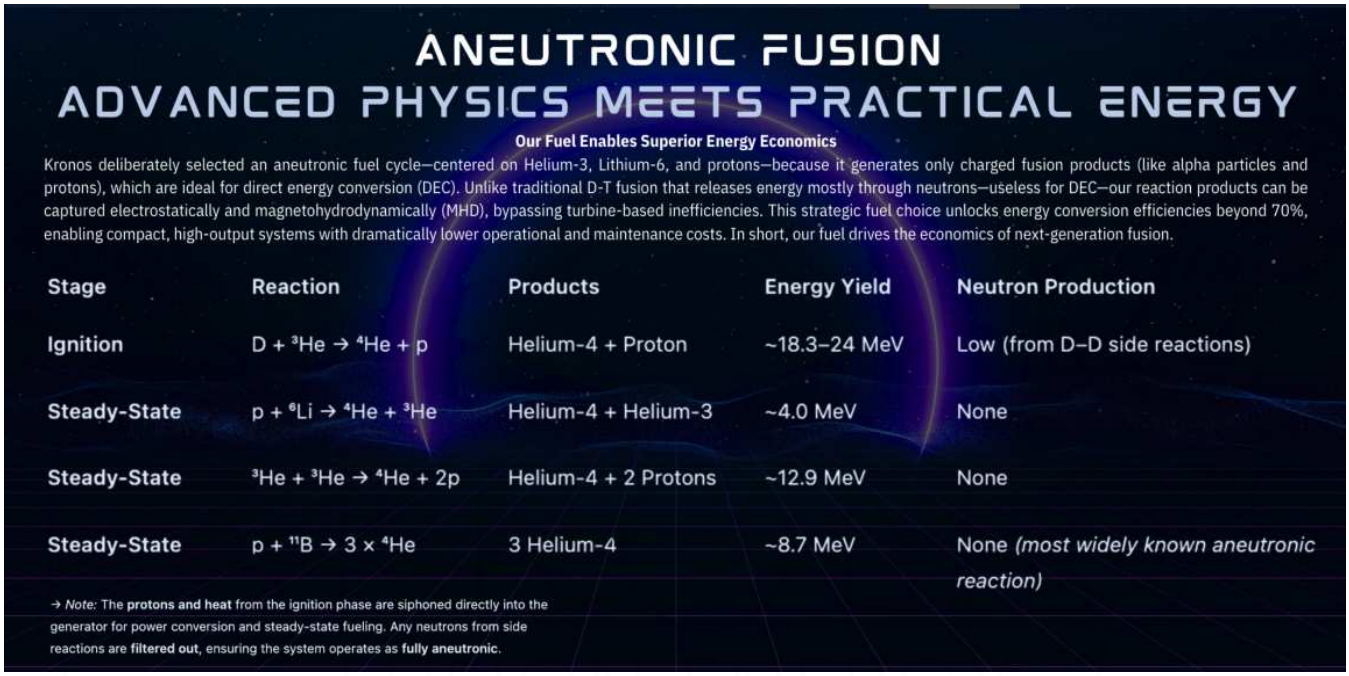


Figure 2: Detailed overview of Kronos' aneutronic fuel cycle strategy, highlighting ignition and steady-state reactions, energy yields, and neutron output. This chart emphasizes the use of helium-3, lithium-6, and proton fuels to enable high-efficiency, neutron-free power generation ideal for direct energy conversion.

The reactor has no need for a thick tritium-breeding blanket or massive radiation shield; consequently, critical components such as the vacuum vessel, magnets, and direct conversion structures can be placed much closer to the plasma. This proximity allows a very compact plant layout and reduces power loss in transmission (for example, magnetic fields and electrostatic fields used for direct conversion can directly interact with the fusion products before they thermalize). The aneutronic approach also means minimal activation of structures, enabling high uptime and easier maintenance or component replacement when needed. In summary, the reactor is *aneutronic by design*: it avoids neutrons at the source to simplify the entire power plant.

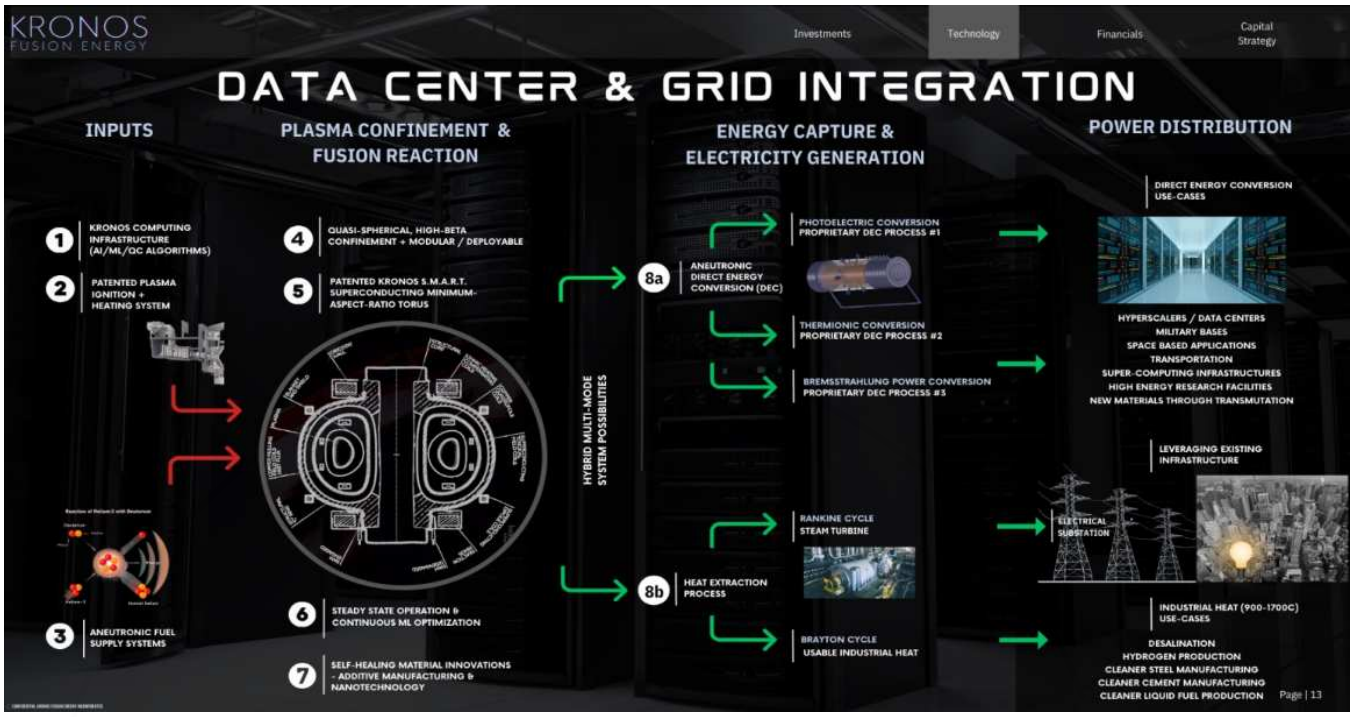


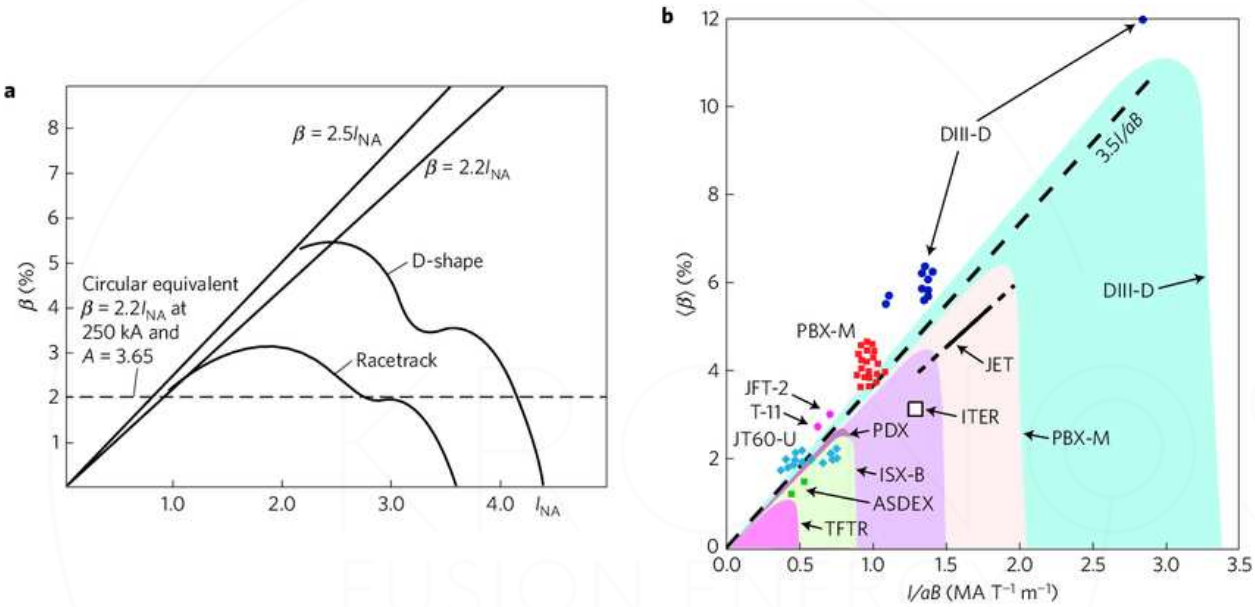
Figure 3: Multi-modal energy capture architecture showcasing MHD generators, electrostatic collectors, and bremsstrahlung photon harvesting systems for high-efficiency output.

Achieving aneutronic fusion conditions necessitates confronting higher temperature and confinement requirements. S.M.A.R.T. addresses this by aggressively pushing the magnetic field strength and optimizing the plasma geometry for stability and high β. The guiding philosophy is to turn what is often seen as a constraint – the limits of HTS magnet strain and plasma stability – into a driver of innovation. The design process started with the materials and physics limits (e.g., the 0.4% strain limit of ReBCO, the need for >100 keV ion temperatures) and then engineered solutions to reach those limits safely. In effect, instead of incrementally improving on existing tokamaks, S.M.A.R.T. is a clean-sheet design that reimagines each subsystem (magnets, structure, plasma heating, power extraction) in light of those extreme targets.

Reactor Geometry and Plasma Configuration: S.M.A.R.T. adopts a spherical tokamak configuration with an ultra-low aspect ratio ($A \approx 2.0$, meaning the major radius R is only about twice the minor radius a). The major radius in our design is $R_0 \approx 2.0$ m and minor radius $a \approx 1.0$ m, giving a plasma volume on the order of $V \approx \pi a^2 R_0 \sim 6–7$ m³. Although smaller in volume than conventional large tokamaks, this plasma is made exceptionally potent by high magnetic field and shaping. The spherical tokamak approach (sometimes called a spherical torus) is known to allow higher plasma beta – that is, a higher ratio of plasma pressure to magnetic pressure – than standard aspect ratio designs. In past experiments, spherical tokamaks have achieved toroidal beta values of 30–40%, several times those in conventional aspect ratio machines. S.M.A.R.T. leverages this property: a high β allows more fusion power for a given magnetic field strength. In fact, an approximate performance scaling for tokamaks is **Performance $\propto \beta^2 B^4 V$** (where B is the magnetic field and V the plasma volume). This scaling underscores that even at fixed field and size, pushing β higher greatly multiplies fusion output, and conversely at fixed β, a higher B multiplies output to the fourth power. S.M.A.R.T.'s spherical design thus seeks to maximize both β and B.

To enhance stability and confinement, the plasma cross-section is highly shaped: we employ a substantial elongation (elongation factor $\kappa > 2.5$) and a negative triangularity ($\delta < 0$, meaning the plasma is D-shaped with the flat side on the inside and the "triangle" pointing outward, opposite to most tokamaks). High elongation increases the plasma

aspect ratio vertically and improves the plasma's self-organized current distribution and magnetic shear, which tends to improve confinement and bootstrap current generation. The negative triangularity is a novel choice informed by recent experiments that show improved edge stability and the avoidance of deleterious edge-localized modes (ELMs) when the plasma has a reversed triangular shape. By having $\delta < 0$, the plasma's outer top/bottom shape flares outward, which allows us to place divertors at larger major radius positions. This geometric change spreads out the exhaust interaction over a larger area and volume, simplifying the divertor heat exhaust challenge and providing more room to insert direct conversion structures in the exhaust stream. Additionally, negative triangularity causes the inboard side of the plasma (the side facing the central column) to be more concave, effectively increasing the clearance around the central column at the midplane. This reduces the magnetic flux density "hot spot" on the inner leg of the toroidal field coil and spreads out forces, which enhances coil structural rigidity and longevity.



Charts 2: Plasma shaping and beta performance. (a) Comparison of normalized beta (β) versus normalized current (I_n/NA) for different plasma cross-sections—circular, D-shape, and racetrack—showing the higher stability limits achievable with optimized shaping. (b) Experimental database of beta versus I_paB across major tokamaks (JET, ITER, DIII-D, PBX-M, ASDEX, etc.), illustrating the path toward advanced high-β operation and confinement scaling.

In summary, plasma shaping in S.M.A.R.T. serves both physics and engineering aims: it improves MHD stability (mitigating instabilities and enabling higher beta) and simultaneously aids in managing the thermal and mechanical loads.

Figure 4: Schematic rendering of the S.M.A.R.T. tokamak architecture showcasing a compact, high-field configuration tailored for aneutronic fusion. This figure hides the actual geometry.

Every dimension of the plasma-facing components and magnets is numerically optimized to work in concert. The design process used computational tools to iterate the plasma geometry so as to minimize localized stress concentrations and electromagnetic loads on structures. For example, the angle of each toroidal field coil leg and the spacing between coils were adjusted to ensure that the J×B forces (Lorentz forces due to current J in magnetic field B) are well-distributed. The resulting geometry is the outcome of an interplay between plasma physics requirements and mechanical constraints: it is compact and high-performance, yet avoids pushing any material beyond its limit.

Magnet Design and Structural Support: At the heart of S.M.A.R.T. is an extremely high-field magnet system. The toroidal field coils are wound with ReBCO high-temperature superconductor tape, chosen for its ability to remain superconducting under very high magnetic fields and at relatively high temperatures (20 K or even up to 77 K for moderate fields, though here we operate at cryogenic temperatures ~4–20 K to maximize performance). The target peak field at the ReBCO conductor is on the order of 30–35 T, which is unprecedented in tokamak magnets (for comparison, the highest-field tokamak under construction, SPARC, projects ~23 T on coil). On the plasma axis at $R \approx 2$ m, the field is about 13–14 T. This high B-field is a major driver of compact fusion power density but creates tremendous electromagnetic forces on the coils. The challenge is to support these forces without exceeding the roughly 0.4% tensile strain limit of the ceramic superconductor.

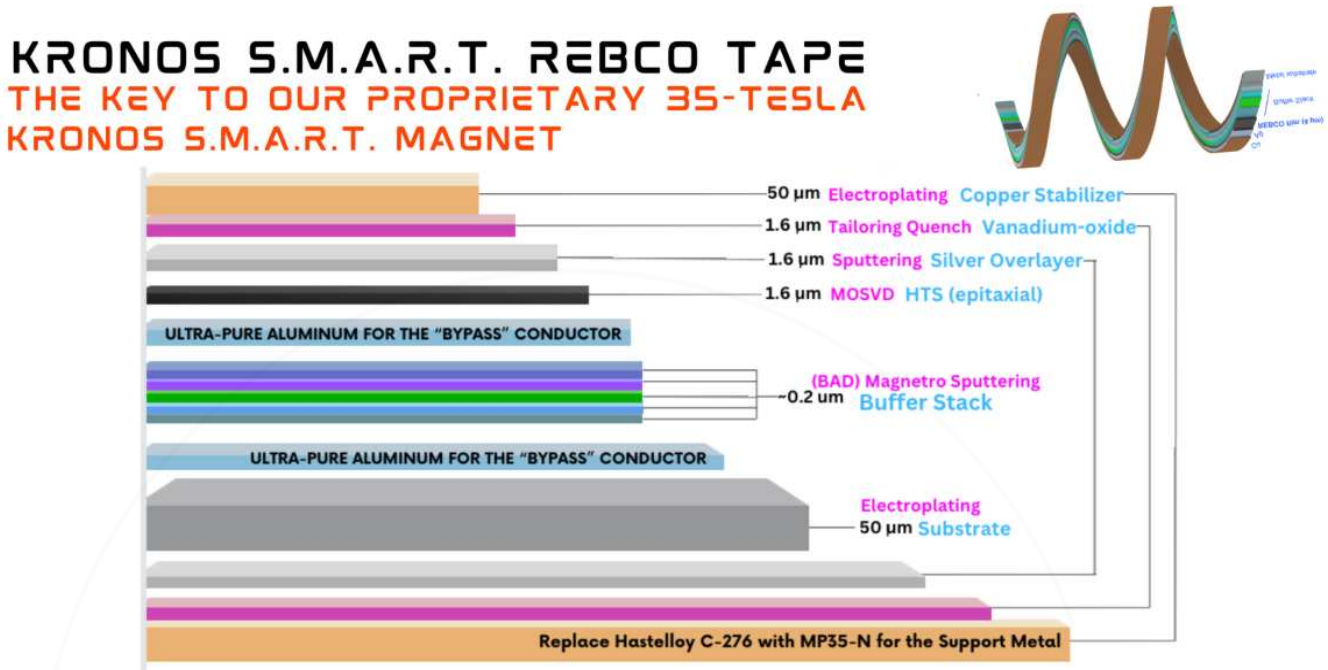
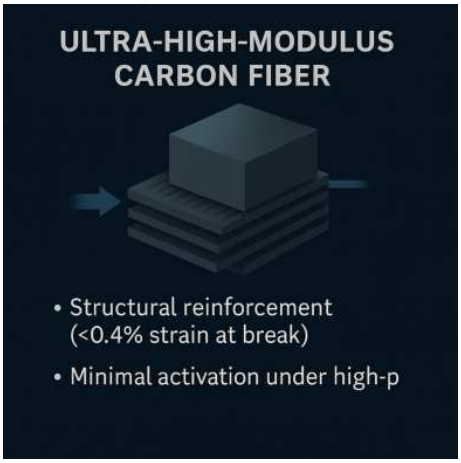


Figure 5: Cross-sectional schematic of the Kronos S.M.A.R.T. REBCO tape (not to scale). Multiple functional layers are integrated: a high-modulus MP35N metal substrate (gray) with textured oxide buffer (white) supports a thin epitaxial REBCO superconductor layer (green). A silver overlayer (light gray) and a tailored vanadium-oxide coating (magenta) cover the HTS. The tape is stabilized by a thick copper layer (orange) and additionally embeds ultra-pure aluminum conductors (blue) as bypass paths. Layer thicknesses are indicated (in microns where µm, and substrate/stabilizer in microns); MP35N replaces the conventional Hastelloy substrate to maintain <0.4% strain in the HTS.

The S.M.A.R.T. magnet system is therefore augmented with an array of structural innovations to manage stress and strain:

High-Modulus Materials: Rather than using conventional coil structural materials (e.g., stainless steel or Hastelloy substrates which have Young's modulus ~200 GPa), S.M.A.R.T. employs materials with exceptionally high elastic modulus to carry the load. The ReBCO tapes are fabricated on a strong, stiff substrate alloy (MP35N, a nickel-cobalt-chromium-molybdenum alloy) which has E ≈ 230 GPa and very high tensile strength (>1.5 GPa yield). By using MP35N instead of a softer steel alloy in the superconductor tape, the tape itself can withstand higher stress before straining the superconductor. Surrounding the coil, we incorporate reinforcing structures made of ultra-high modulus carbon fiber composites (e.g., Toray M40J/M46J fibers, with E ~ 400 GPa). These carbon fiber reinforced polymer (CFRP) components have strength in excess of 4 GPa and can sustain huge loads with minimal elongation. They are strategically used in coil clamps, supports, and even as an external wrapping ("cocoon," described below). By ensuring all key structural members have modulus in the hundreds of GPa, the system can support magnetic pressures on the order of multiple GPa without significant deformation – thus keeping the ReBCO strain under the 0.4% limit. In essence, we design the magnet support structure to be as stiff as possible, so that the superconductor is never stretched beyond its safe elastic range.



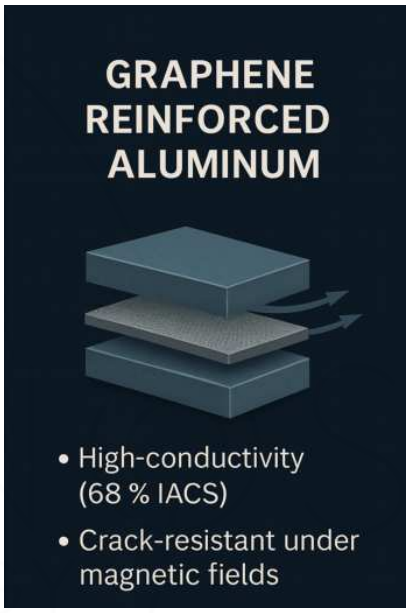
Bucking Cylinder: To counteract the immense inward "centering" force that the toroidal field coils exert on each other (tending to crush the central column), S.M.A.R.T. uses a rigid ceramic bucking cylinder at the center of the tokamak. This cylindrical structural element (made of a high-compressive-strength, non-conductive ceramic or composite) runs inside or alongside the inner legs of the TF coils and takes up the compressive load that would otherwise squeeze the coil bundle. By having a dedicated bucking post, we relieve the coil's own structural material from having to bear all the inward force, preventing the HTS tapes from being crushed or delaminated. Finite element analysis has shown that this measure drastically reduces the peak compressive stress in the coil windings.

Force-Balancing "Cocoon": The entire magnet assembly (all TF coils and support structures) is enclosed in a continuous external compression shell – informally called a carbon-fiber "cocoon." This shell is wound or assembled under tension such that it applies a uniform inward pressure on the magnet system even before the tokamak is energized. It pre-compresses the coils, keeping them in a state of compressive stress at zero operational load. When the coils are energized and try to push outward or deform under Lorentz forces, they first have to overcome this preset compression. Effectively, the cocoon ensures that the coils never experience net tension beyond a small margin, because the worst-case Lorentz tensile forces only relieve some of the pre-compression instead of pulling the material apart. This approach of pre-stressing is analogous to prestressed concrete in architecture – here applied via high-strength fiber composites. It also helps damp vibrations and cyclic fatigue; the snugly fitting shell prevents the coils from rattling or accumulating cyclical strain that could lead to cracks over many pulses.

Ice-Powered Support Pistons: One of the more unique features of the S.M.A.R.T. support structure is the use of cryogenically actuated "ice pistons" to counter vertical forces. The TF coils form a D-shaped loop, and when current flows, the top and bottom sections of the D experience strong vertical forces (similar to an outward bowing or an "arching" force trying to straighten the D). These vertical Lorentz forces can cause the coil to stretch upward at the midplane, concentrating stress at the inboard midplane ("throat") region. To mitigate this, we place special pistons or jacks that are filled with a substance (for example, a water/gas mixture) that is liquid during assembly and then frozen solid during operation. When the substance freezes, it expands (water, for instance, expands ~9% upon freezing to ice). This expansion is used to push against the coil supports, effectively jacking up and pre-loading the structure in the opposite direction of the Lorentz force. The result is that when the magnet is energized, the additional upward force finds the structure already braced against it. The ice pistons thus serve as passive, self-regulating actuators that limit vertical displacement and strain in the coils. They operate without active control – the phase change is triggered by the cryogenic environment of the magnet – making them simple and reliable. By reducing vertical strain in the coil, we further ensure the ReBCO tapes in the inner leg remain safely within stress limits.

Helical Coil Pitch: The toroidal field coils in S.M.A.R.T. are not perfectly planar; each coil winding is given a slight helical twist (on the order of a few degrees across the height of the machine). This intentional coil pitch means that the Lorentz forces, which would normally press the coils directly radially and vertically, are re-distributed along a helical path. In effect, the forces that would cause the coil to bend or overturn are spread out, reducing peak stresses on any single structural element. Additionally, the small helical component helps the toroidal field coils contribute a bit to poloidal field, blurring the line between TF and poloidal field (PF) coils. This "hybridization" allows some load sharing and can reduce the requirement on separate PF coils for equilibrium, thereby simplifying the mechanical loading. Calculations show that a ~3–5° coil pitch can reduce the overturning moment and peak support stress by roughly 50%. It is a subtle design tweak that significantly eases mechanical stress.

Non-Insulated HTS Windings: S.M.A.R.T. adopts a no-insulation winding technique for the HTS coils. In conventional magnets, each turn of the superconductor is insulated from the next to prevent shorting; however, this means that if a local region quenches (loses superconductivity), the current in that turn cannot go anywhere and a large voltage builds up, potentially causing damage or quench of the entire coil. In a non-insulated coil, the turns are directly in electrical contact (or separated by only a very thin oxide layer that allows some current bleed-through). If a small section of one turn becomes resistive, the current can divert into adjacent turns and bypass the "hot spot," effectively self-healing the quench by spreading the current. This vastly improves quench protection and can allow the magnet to recover without a full system shutdown. The tradeoff is a small amount of circulating current between turns and slightly slower field ramp, but these are acceptable in a steady-state machine. We enhance this concept by using a *smart insulation* coating: a thin layer of vanadium oxide (V_2O_3) on the tape, which has the property of increasing resistivity at low temperature. This provides just enough inter-turn resistance to prevent excessively large eddy currents during transients, while still permitting quench current sharing when needed. The NI approach mitigates the risk of sudden magnet failure and also reduces the need for bulky external dump resistors or complex quench detection systems, since the coil intrinsically limits voltage spikes. Demonstrations by other research groups have shown that NI HTS coils can survive localized heating without damage, a property S.M.A.R.T. heavily relies on for safe high-field operation.



Conductive Bypass and Thermal Management: In addition to structural reinforcement, the magnet system includes a high-conductivity bypass layer made of graphene-enhanced pure aluminum. This thin aluminum lamination within or around the HTS tape serves multiple purposes. First, aluminum at cryogenic temperatures has extremely high thermal conductivity, which helps quickly spread any localized heat (from a small quench or external source) across the coil, preventing hot spots. Second, aluminum provides a low-resistance path for current to redistribute in the event of a quench (complementing the NI strategy) and can carry current temporarily if a section of superconductor drops out. By reinforcing this aluminum with graphene (which has exceptional thermal and electrical properties), we improve its performance in the presence of strong magnetic fields – pure aluminum can suffer increased resistivity under fields (magnetoresistance), but the graphene reduces this effect. The net result is a coil that is not only mechanically robust but also electrically and thermally stabilized against disturbances.

Through this combination of high-strength materials, geometric tailoring, and novel supports, the S.M.A.R.T. TF coils are expected to sustain the target ~30 T fields with a safety margin on strain. Structural analysis indicates that the peak combined stress in the coils (on the order of 1.5–1.6 GPa) will translate to ~0.4% strain in the ReBCO – essentially at the design limit but not beyond. The factor of safety is kept modest to maximize performance, but all components are engineered for fatigue endurance so that even after thousands of pulses (or years of steady operation), no cumulative damage like cracking or delamination should occur. Notably, because the reactor produces negligible neutrons, the magnet materials will not suffer radiation embrittlement or swelling, giving them a long service life compared to magnets in a D–T machine.

Heating & Current Drive				
Kronos Design Feature	Value Proposition / Innovation	Quantitative Performance	Reference Institution	Peer-Reviewed Source
Neutral Beam Injection (NBI)	Heats and drives current with high-energy ions	JT-60U achieved 500 keV beams, 10 MW class	JAEA	JT-60U NBI Program
Radiofrequency Heating (ECRH/ICRH)	Provides localized plasma heating and current drive	ITER: 20–67 MW RF planned	ITER	ITER Heating Systems
Multi-Modal Heating	Redundancy ensures reliable ignition and sustainment	ASDEX Upgrade uses NBI + ICRH + ECRH simultaneously	Max Planck IPP	ASDEX Heating Systems
Long-Pulse Superconducting Systems	Sustains steady-state plasmas for minutes to hours	EAST sustained >100 s plasma with multi-MW heating	EAST (CAS)	EAST Tokamak Results

Table 1: Kronos follows the proven path of JT-60U, ITER, ASDEX, and EAST—deploying redundant NBI + RF systems to reach and sustain aneutronic ignition regimes.

Plasma Heating and Current Drive: Initiating and sustaining an aneutronic plasma requires achieving much higher temperatures than in a D–T reactor. For p–^{11}B fusion, for instance, ion temperatures on the order of 100–200 keV (up to ~billion degrees) are desirable for significant reactivity, whereas D–T can ignite around 10–20 keV. S.M.A.R.T. employs multiple layers of heating and current drive to reach and maintain these conditions:

Dual-Stage Ohmic Heating (OH) Solenoids: Rather than a single central solenoid, we use two concentric ohmic heating coil systems. The first (OH-1) is a fast-ramp, high-voltage solenoid designed to induce a quick current spike in the plasma for initial breakdown. It is made with a specialized winding (for example, Litz wire or twisted HTS) that can sustain a high dI/dt, creating a large loop voltage (>1 kV) momentarily to ionize the gas and initiate the plasma with minimal delay. However, OH-1 is not optimized for long pulses – it trades current capacity for speed. The second solenoid (OH-2) is a high-current, high-flux coil that provides the bulk of the magnetic flux change needed for driving plasma current over a longer period. OH-2 has a much larger volt-second capability and can drive the plasma current for an extended duration or even support steady-state operation in concert with other drives. By separating the OH function into two, we avoid the compromise where one coil would be suboptimal for either startup or sustainment. In effect, OH-1 handles the "spark" and OH-2 handles the "burn." Once the plasma is ignited and other current drive kicks in, OH-2 can taper off its assist to avoid excessive resistive losses.

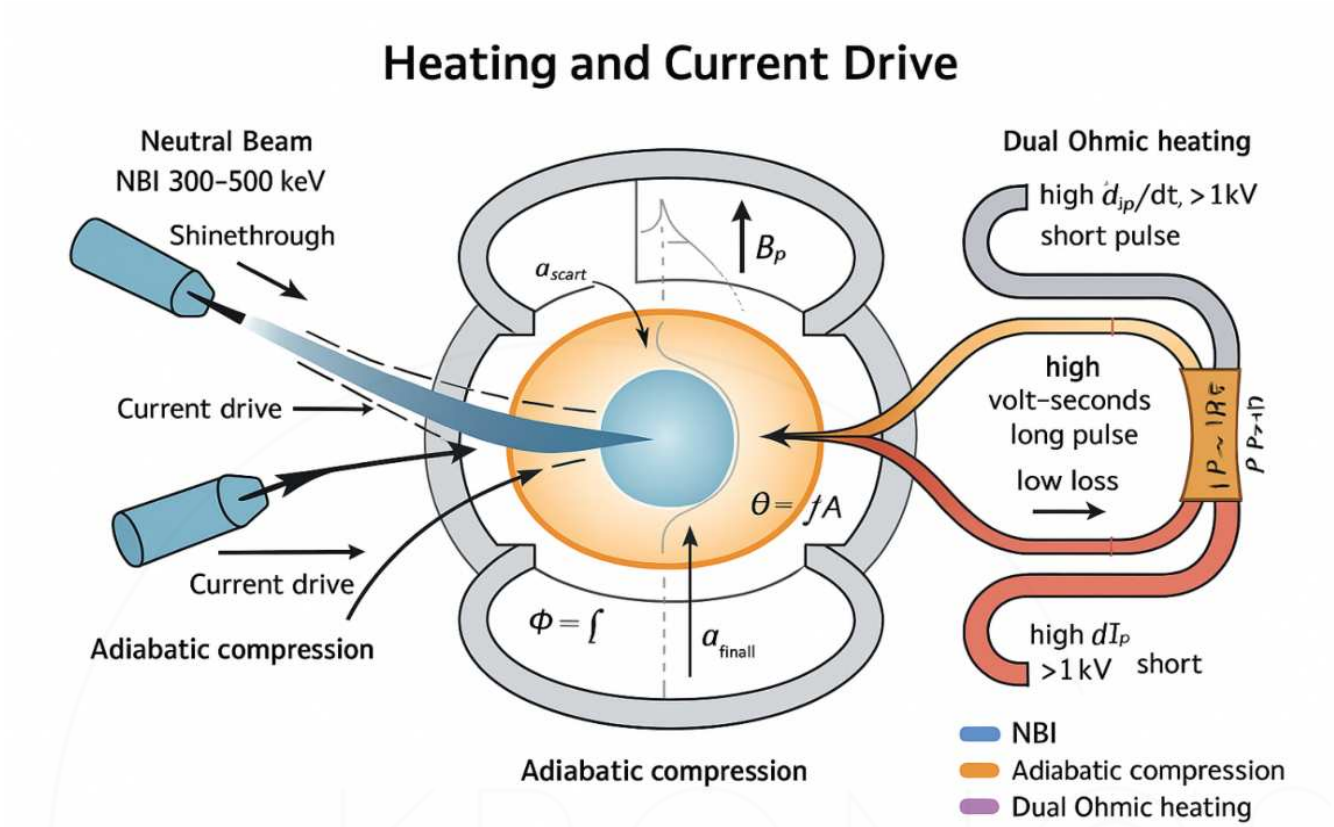


Figure 6: Schematic of a closed-loop Brayton cycle. Heated working fluid enters the turbine at 1020 °F (high pressure), expands to generate power, and exits at 800 °F (low pressure). The recuperator transfers heat between the hot and cold streams, improving efficiency. The fluid is then cooled to 110 °F at low pressure before being compressed back to 240 °F at high pressure, completing the cycle.

Adopting dual OH systems ensures flexibility: we can tailor the current ramp-up precisely with OH-1 and then maintain or adjust the current smoothly with OH-2. Using two systems is admittedly more complex in engineering (two sets of coils and power supplies), but it provides a robust solution to the conflicting requirements of breakdown vs steady drive. If ongoing R&D finds ways to significantly improve the current density of fast-ramp HTS coils, it might be possible to simplify to a single OH system, but with current technology the dual approach is prudent.

Adiabatic Compression: A distinctive method to boost plasma temperature and current in S.M.A.R.T. is through radial compression of the plasma column. After the plasma is ignited at a relatively larger minor radius (slightly outboard of the chamber center), the entire plasma is shifted inward by active control coils or shaping field changes. As the plasma moves into a region of higher field (closer to the toroidal coil inner leg) and a smaller volume, the magnetic flux through the plasma cross-section is conserved. By compressing the plasma inward (and possibly shrinking its cross-sectional area slightly), the toroidal field experienced by the plasma increases and so does the plasma current (via flux conservation and the pinch effect). This adiabatic compression process can ramp up the plasma current and heat the plasma (through magnetic compression heating) without additional input power – essentially trading magnetic field energy for plasma thermal energy. It's analogous to compressing a gas to heat it, but here compression is magnetic/geometric. The result is higher temperature and density, aiding the approach to aneutronic fusion conditions. This technique also reduces demands on the ohmic solenoid, as the final plasma current can be higher than what the solenoid alone initially induced.

Key Specifications and Expected Performance of the Heating/Current-Drive Subsystems in Kronos S.M.A.R.T.				
Subsystem	Role & Key Specs	Performance Benchmark	Technology Readiness (TRL)	Peer-Reviewed Source
OH-1: Fast Ohmic coil	Provides high-voltage (~1 kV loop) for rapid breakdown; Litz-wire or twisted HTS conductor; short pulse (~10–50 ms) capacity.	Achieves initial plasma E~1 V/m field for breakdown (e.g. 0.7 m radius, 7 V loop). Designed for >100 kA start current in <5 ms.	~4 (Prototype stage; high-voltage HTS pulsed coils not yet used in reactors).	Fusion Economics Report (2021)
OH-2: Sustaining Ohmic coil	Drives long-pulse current; high volt-second (>10 Wb) superconducting solenoid (ReBCO); high J capacity with low resistive losses.	Capable of sustaining multi-MA plasma current for several seconds (e.g. 1–2 MA for >5 s). Central solenoids used in ITER-class devices achieve >15 MA for >300 s with Nb_3Sn. ReBCO coil operation demonstrated at 20 T (MIT/SPARC prototype).	~5 (Component validation; large HTS solenoids under development, smaller copper/NbTi solenoids TRL 9).	Economics of Fusion Power (2016)
Adiabatic Compression	Rapid magnetic compression of plasma (reduce R by ~20–30%, increase B by ~20–50%); boosts I_p, T, n post-formation.	Doubling of I_p and significant T_e rise observed in ST experiments (e.g. START, MAST) when R halved. Expected to raise S.M.A.R.T. T_i from ~100 keV to 200 keV post-compression.	~3 (Proof-of-concept in experiments; not yet implemented in power-reactor environment).	STEP Costing Report

Neutral Beam Injection	High-energy neutral beams (H or D) for core heating and current drive; 300–500 keV beam energy, total power 10–20 MW; multi-path tangential injectors.	500 keV deuterium beams achieved on JT-60U; $\sim$20 MW NB systems routine on large tokamaks (JET, JT-60). S.M.A.R.T. NBI expected to heat plasma to >100 keV and drive >0.3 MA non-inductively.	~7 (High-energy NB technology demonstrated in large physics devices; needs integration in compact reactor setting).	Fusion Industry Association Report (2023)

Table 2: Summary of S.M.A.R.T. heating and current-drive subsystems, with roles, key parameters, example benchmarks from experiments or designs, and approximate Technology Readiness Level. TRL is assessed on a scale of 1–9 (with 9 being state-of-the-art in operational use).

Adiabatic compression has been used in some form in certain tokamak operations (e.g., lowering plasma inductance), but S.M.A.R.T. would utilize it deliberately as a heating/current boost mechanism. The plasma control system would be programmed to form the plasma slightly off-axis and then bring it in rapidly once a certain current/temperature threshold is reached, timing it to maximize the benefit without causing instability. The compression ratio is limited by how much the plasma can move and compress without losing confinement, but even a moderate inward shift could raise B and I by ~20–30%, which directly increases fusion power and current drive.

Neutral Beam Injection (NBI): For continued heating and non-inductive current drive, S.M.A.R.T. incorporates high-energy neutral beam injectors. Our design calls for one or more NBI systems capable of delivering neutrals in the 300–500 keV energy range. These beams consist of accelerated ions (likely hydrogen or deuterium, or possibly ^{3}He depending on scenario) which are neutralized and then injected tangentially into the plasma. When these fast neutrals penetrate the plasma, they ionize and transfer their energy to the plasma particles via collisions, rapidly heating the plasma (especially the ions). Additionally, because the beams are aimed tangentially, the momentum of injected particles drives plasma rotation and generates plasma current (beam-driven current). In the spherical tokamak geometry, designing the NBI is tricky due to the compact space and high plasma density, but tangential injection at the mid-plane can maximize penetration depth and core heating. We tailor the NBI energy to be high enough to avoid being stopped too early in the plasma (ensuring core heating) but still manageable in size and power requirements. High-efficiency neutralizers and energy recovery from un-neutralized ions are included to keep the overall electrical efficiency of the NBI system high, as aneutronic operation demands minimal wasted input power. The NBIs are critical for supplementing ohmic heating, especially since ohmic heating effectiveness diminishes at high temperature (plasma resistance drops). They also help sustain a substantial fraction of the plasma current in steady state, together with the plasma's self-driven bootstrap current.

Neutral beams have been a workhorse in tokamaks, and their physics is well-understood. For S.M.A.R.T., one consideration is that the direct conversion systems (like MHD channels or electric grids) should not interfere with beam access; thus, we allocate distinct beam ports that bypass the DEC structures. The beamline is also designed to inject slightly downward (or upward) to align with the shaped plasma and avoid hitting the wall. The beam parameters can be adjusted based on the fuel scenario – e.g., a deuterium beam might be used even in a mostly proton/helium plasma just as a heating tool, since its deposition and current drive characteristics might be favorable.

Bootstrap Current Optimization: The plasma shaping and high β operation in S.M.A.R.T. naturally encourage a large bootstrap current – a self-generated toroidal current that arises from pressure gradients in the plasma. Bootstrap current can significantly reduce the need for external current drive. By operating in regimes of high normalized

pressure and with appropriate pressure profiles, our design expects a substantial fraction of the needed plasma current to come from the bootstrap effect (potentially on the order of 50% or more of total current). The high elongation and negative triangularity are, in part, intended to maximize this effect while maintaining stability. During steady operation, the combination of bootstrap current plus NBI-driven current (and perhaps a small amount of RF or other drive if needed) should maintain the plasma indefinitely without relying on the ohmic transformer, enabling true steady-state operation – a necessity for a power plant.

Each of these subsystems operates semi-independently, providing multiple layers of redundancy and flexibility. For example, if the NBI needs to be turned off for maintenance, the ohmic and compression can handle shorter operation, or if the ohmic system saturates, the NBI and bootstrap can carry more of the load. This modularity enhances reliability. In summary, the heating and current drive architecture of S.M.A.R.T. is tailored to produce and sustain the extreme plasma conditions required for aneutronic fusion, with innovative techniques like plasma compression and dual ohmics augmenting well-proven methods like NBI and bootstrap utilization.

Energy Conversion (Direct + Hybrid)				
Kronos Design Feature	Value Proposition / Innovation	Quantitative Performance	Reference Institution	Peer-Reviewed Source
Direct Charged-Particle Conversion	Captures kinetic energy of alphas directly into electricity	~48% efficiency in Tandem Mirror Experiment	LLNL	Moir & Barr, Nucl. Fusion 1985
Hybrid Direct + Brayton Cycle	Raises total efficiency by combining DEC with thermal loop	~49% net electric efficiency modeled for D–^{3}He	UW–Madison	Kulcinski et al., Fusion Tech. 1992
Magnetohydrodynamic Conversion	Harvests plasma exhaust before thermalization	Adds ~10% recovery in aneutronic concepts	NASA JSC	NASA MHD Studies (AIAA 1991)
Photon (Bremsstrahlung) Capture	Converts plasma X-ray emission via photovoltaics	X-ray direct conversion ~20% in experiments	LANL / LBL	Hard X-ray Conversion Efficiency (Nature Comm.)

Table 3: Kronos blends proven direct converters (48% at LLNL) with high-T Brayton loops, plus MHD and radiovoltaics, pushing efficiency well beyond conventional steam cycles.

Direct Energy Conversion Systems: A cornerstone of the S.M.A.R.T. architecture is the integration of direct energy conversion (DEC) systems to capture fusion energy in forms other than heat. In an aneutronic reactor, because the fusion products are charged particles and high-energy photons, it becomes feasible to convert their energy to electricity *before* they randomize into heat.

NEUTRONIC FUSION FUEL CYCLES

Reaction		Products (MeV)	E_{Out} (MeV)
$D^2 + D^2$	$\xrightarrow{50\%}$	$T(1.01) + p(3.02)$	4.0
	$\xrightarrow{50\%}$	$He^3(0.82) + n(2.45)$	3.3
$D^2 + T^3$	$\longrightarrow$	$He^4(3.5) + n(14.1)$	17.6
$p + Li^7$	$\xrightarrow{20\%}$	$2He^4 + (17.3)$	17.3
	$\xrightarrow{80\%}$	$Be^7 + n(-1.6)$	-1.6

30% EFFICIENT, THERMAL CYCLE

ANEUTRONIC FUSION FUEL CYCLES

Reaction		Products (MeV)	E_{Out} (MeV)
$p + Li^6$	$\longrightarrow$	$He^4(1.7) + He^3(2.3)$	4.0
$p + B^{11}$	$\longrightarrow$	$3He^4(8.7)$	8.7
$p + N^{15}$	$\longrightarrow$	$He^4(3.7) + C^{12}(1.2)$	4.9
$D^2 + He^3$	$\longrightarrow$	$He^4(3.6) + p(14.7)$	18.3
$D^2 + Li^6$	$\longrightarrow$	$2He^4\ (22.4)$	22.4
$He^3 + Li^6$	$\longrightarrow$	$2He^4(1.9) + p(15.0)$	16.9
$He^3 + He^3$	$\longrightarrow$	$He^4(4.3) + 2p(8.6)$	12.9

90% EFFICIENT, THERMAL CYCLE

Charts 3: Comparison of D–T versus aneutronic fuel cycles, highlighting reduced neutron production and enabling direct energy conversion pathways in the S.M.A.R.T. design.

The S.M.A.R.T. design incorporates several complementary DEC pathways, each targeting a different component of the fusion output:

Magnetohydrodynamic (MHD) Energy Extraction: Surrounding the plasma or placed in the plasma exhaust stream, we channel ionized fusion exhaust gas through magnetic ducts to produce electrical power. Essentially, as charged particles (like 3.6 MeV alphas or 14 MeV protons from p–^{11}B) move outwards, they pass through strong magnetic fields in specially designed channels. The motion of these charges in B-fields induces electric currents in electrode structures (via the v×B force on charges), and we capture this induced current with electrodes and closed circuits – functioning like an MHD generator. Because S.M.A.R.T.'s TF coils generate such high fields (20–30 T even near the plasma periphery), the MHD channels can be compact yet still induce substantial electromotive force. The plasma exhaust, guided by divertor magnetic fields, is directed through these channels, where electrically conductive plates or loops pick up the current and feed it out as DC power. This approach directly harvests the kinetic energy of the charged fusion products (and plasma flow) without intermediate thermalization, thereby bypassing the inefficiencies of a thermal cycle. The MHD conversion efficiency improves with higher conductivity of the plasma and higher magnetic field; our high-B design is ideal from that standpoint. One challenge is that the plasma in the exhaust must remain sufficiently ionized and free of insulating debris for MHD to work well – which suggests maintaining a modest plasma temperature in the divertor leg or seeding a low-ionization-point gas to improve conductivity. We have factored in active control of divertor plasma conditions to optimize MHD output. MHD generation in fusion has precedent (experiments in the 1980s and 90s achieved significant direct conversion fractions in mirror machines), and S.M.A.R.T. aims to implement it at power-plant scale.

Electrostatic Direct Collectors: One of the most straightforward direct conversion techniques for charged particles is electrostatic deceleration and collection. We equip the reactor with a series of biased electrodes or grids that create an electrostatic potential field around the plasma periphery and in the exhaust ducts. As positively charged fusion products (alphas, protons) escape the confining magnetic fields, they encounter these electric fields which decelerate them. By setting the electrode potentials appropriately (on the order of the charged particle energy, e.g., megavolts), the kinetic energy of the particles is converted into electrical potential energy. The charged particles then deposit on a collector plate, delivering their charge at high potential. In effect, the fusion ions perform work on the electric field and generate a DC current through an external circuit connected to the collectors. In S.M.A.R.T., we consider a multi-stage deceleration, where particles might pass through several rings or grids with decreasing

potential to minimize energy loss as radiation (spreading out the deceleration prevents too many X-rays from sudden braking). The final collector could be, for instance, a negatively biased plate that the ions come to rest on, having given up most of their energy. This method can be extremely efficient – theoretical efficiencies of 80–90% are possible for converting kinetic energy if space-charge effects and grid interception losses are managed. Practically, achieving these numbers requires fine-tuned vacuum and field configurations so that the ions don't scatter or prematurely neutralize. We plan to implement electrostatic converters in regions just outside the main plasma (for charge escaping during normal operation) and in the direct line of the divertor outflow (which contains a lot of fast charged particles post-reconnection or mirror escape). Because our reactor avoids neutrons, we can use high-voltage structures much closer to the plasma without them being quickly damaged or activated, which is a major advantage of our approach. The technology for such direct converters was explored in concepts like the Lawrence Livermore "Venetian blind" converter – we build on those ideas with modern materials and higher fields.

Thermionic and Thermoelectric Conversion: Some of the fusion energy will inevitably thermalize in material structures (for instance, plasma-facing surfaces will be heated by particle bombardment and any neutral or thermalized flux). To reclaim energy from these surfaces beyond what a coolant loop would do, S.M.A.R.T. integrates advanced thermionic converters. A thermionic converter consists of a hot electron-emitting surface and a cooler collector in vacuum; the hot surface (emitter) releases electrons due to thermionic emission, and those electrons are collected by the cooler surface, establishing a current. By coating plasma-facing components (first wall panels, divertor plates) with low work-function materials like lanthanum hexaboride (LaB_6) or cesium-embedded surfaces, we enable them to emit electrons at elevated temperatures. For example, divertor plates in our design might operate at 1000–1500 °C due to absorbed heat; at those temperatures, LaB_6 can emit copious electrons. An opposing electrode at a slightly lower temperature or with a positive bias collects the electrons, generating power. This direct conversion of heat to electricity bypasses the need for fluid coolant in those zones and works even in vacuum or microgravity (which is why thermionics have been considered for space power systems). The efficiency of thermionic conversion can be 10–20% for practical temperature ranges, so this is a secondary contributor, but it reduces the heat that needs to be carried away. Similarly, we can employ solid-state thermoelectric modules behind the first wall to convert temperature gradients to electricity (these use the Seebeck effect). We envisage using advanced high-temperature thermoelectrics or thermionic-semiconductor hybrids that can survive fusion wall conditions. These systems are small and static (no moving parts), adding to reliability. In specialized applications like space, the thermionic or thermoelectric output might be the primary mode (since fluid cooling is hard), whereas in a plant, it just augments the total efficiency.

Photonic Energy Conversion: Aneutronic plasmas at high temperature will emit significant electromagnetic radiation, particularly bremsstrahlung X-rays (from electron-ion collisions at tens to hundreds of keV) and synchrotron radiation (from electrons spiraling in the strong magnetic fields). Instead of treating this radiation as pure loss, S.M.A.R.T. captures a portion of it using specialized photovoltaic (PV) systems tuned for high-energy photons. Multi-junction or layered semiconductor devices can directly convert high-frequency photons into electrical current. For instance, we incorporate gallium arsenide (GaAs) and perovskite-based photovoltaic films behind thin radiation-transparent shields in the reactor wall. These "X-ray solar cells" can generate electron-hole pairs from incident X-ray and extreme UV photons. To enhance absorption of these photons (which normally would pass through thin PV cells), we embed high-Z nanoparticles in the PV matrix or use scintillating layers that convert X-rays to visible light, which is then picked up by conventional solar cells. In our design, the interior surface of the outer wall (just behind the plasma-facing layer and any thermionic layer) is lined with these photon-to-electric converters. Bremsstrahlung radiation in aneutronic fusion is significant – it could be on the order of 10–30% of total power in p–^{11}B plasmas – so even capturing part of it directly improves net efficiency. The remainder of the photons that aren't converted will be absorbed as heat in the blanket and can then be dealt with by the thermal cycle, so energy isn't lost. Photonic conversion is a cutting-edge concept; we base our implementation on recent research in direct X-ray detection and conversion materials. While efficiency in turning hard X-rays to electricity is currently

modest, even a 10% efficient capture of the bremsstrahlung would equate to a large power gain in our high-power-density core. As semiconductor and nanomaterial technologies evolve, this number could improve. Importantly, because S.M.A.R.T. places functional materials so close to the plasma (enabled by the lack of neutron destruction), any incremental improvement in these conversion layers directly boosts output.

Secondary Thermal Cycle: After all the direct pathways, any remaining energy that ends up as heat (for instance, the low-grade heat in blanket coolant, or resistive losses, or incomplete particle capture) is extracted via a conventional thermal cycle. We design the reactor's blanket and coolant loops to drive a high-temperature Brayton cycle (using helium or supercritical CO_2 as the working fluid) to generate electricity from heat at an efficiency of ~40–50%. Because the direct conversion has already taken a large portion of the energy (preferably the majority), the thermal cycle can be smaller and more specialized than in a pure thermal reactor. Essentially it acts as a "bottoming cycle" to polish off the remaining energy. In a grid power deployment, this might involve a gas turbine that runs on the hot gas from the blanket. For a smaller or mobile deployment, the thermal system could be a closed-loop radiator or thermionic heat engine. The key is flexibility: S.M.A.R.T.'s energy conversion architecture is hybrid and can be tuned to the use-case. For example, for a deep-space propulsion unit, one might emphasize direct kinetic (MHD) and electrostatic conversion to produce direct plasma thrust or electricity for ion drives, whereas for a terrestrial power plant, one might incorporate a robust Brayton turbine to get every last joule as electricity.

All these conversion subsystems are integrated into the reactor's structure as layered elements. For example, a typical segment of the reactor wall would include: a plasma-facing armor (e.g., HfC-based ceramic) to withstand heat and erosion, backed by a thermionic emitter layer (e.g., LaB_6 on a refractory substrate) to convert heat to electrons, then a coolant channel network for the Brayton cycle (to remove leftover heat and maintain wall temperature), and finally a solid-state PV layer to capture escaping photons. Similarly, the divertor has MHD electrode assemblies in the throats and maybe biased plates for electrostatic capture, with thermionic surfaces on the plate facing the plasma. Designing these layers to function together without interference is an engineering challenge, but our approach benefits from the fact that without neutrons, we can pack these systems in tight and not worry about rapid material activation or degradation – we only manage heat and charged particle flux, which are easier to shield against locally.

The result of this integrated DEC approach is a multi-path energy system capable of exceeding 60% net electrical efficiency, with essentially zero reliance on neutron-driven thermal processes. Every watt produced in the fusion core is either directly converted to electricity or ends up as heat that is then converted at the back end, so that very little is truly "wasted." Improving thermal efficiency is a major goal: every joule not converted to electricity is a joule that must be dumped as waste heat, which in a power plant means bigger cooling systems and lost economic value. By doubling the efficiency relative to a conventional design, S.M.A.R.T. reduces waste heat by a factor of ~3 for the same electrical output. For instance, if a 300 MW_e output requires 1000 MW of fusion power at 30% efficiency (typical of a steam-cycle D–T plant), that leaves 700 MW of heat to get rid of. In S.M.A.R.T., 300 MW_e out of ~500 MW fusion power (60% efficiency) leaves only 200 MW of waste heat, a dramatically lower cooling requirement. This has ripple effects: smaller heat exchangers, less thermal stress on components, easier siting (less warm water discharge or cooling towers needed), etc.

In summary, S.M.A.R.T. treats power conversion as an equal partner to plasma confinement. The reactor structure is not just containing the fusion reaction; it is actively harvesting energy from it at every stage. This holistic approach

distinguishes it from legacy fusion experiments, which often postpone conversion considerations. By building DEC into the architecture, we aim to deliver a fusion generator that produces electricity with an unprecedented combination of efficiency and compactness. The next section will present quantitative expectations for S.M.A.R.T.'s performance based on this methodology, and a subsequent discussion will examine the practicality and challenges of implementing these innovations compared to conventional designs.

KRONOS
FUSION ENERGY

Results

Based on our design studies and scaling analyses, a S.M.A.R.T.-class reactor is expected to achieve performance parameters that are unprecedented for a fusion device of its size. We outline the key anticipated metrics and characteristics:

Magnetic Field and Plasma Current: The toroidal field at the plasma major radius (2 m) is ~13.5 T, with a peak field on the HTS coils of approximately 30–35 T. This extreme field strength, made possible by REBCO superconductors, produces a very high magnetic pressure ($B^2/2\mu_0$) that allows high plasma pressure in a compact volume. The plasma is designed to carry a toroidal current on the order of several mega-amperes; a representative operating point is in the 8–10 MA range of plasma current. This high current, combined with the strong field, gives excellent confinement and stability (high safety factor q at edge due to high shear). The combination of high B and high I also implies a large poloidal field and magnetic shear, beneficial for confinement. For comparison, conventional tokamaks of similar major radius (like JET or SPARC) operate with plasma currents of ~3–8 MA but at much lower field (3–12 T on axis), whereas S.M.A.R.T. drives comparable or higher current at a field an order of magnitude greater than JET's – a regime never achieved before. The high aspect ratio coils in S.M.A.R.T. will produce a steep radial field gradient, but our plasma shaping and position control are designed to keep the plasma precisely centered to avoid instabilities or wall contact.

Plasma Pressure and Beta: A key figure of merit is the plasma beta (β), the ratio of plasma pressure to magnetic pressure. We anticipate operating at a plasma beta (toroidal) in the range of 20–30%. At an on-axis field of ~13 T, this corresponds to extremely high absolute plasma pressures (tens of atmospheres). In practice, a β_t ~ 25% might be realized with volume-average plasma pressure on the order of 1–2 MPa (which is huge, but the magnetic pressure at 13 T is ~270 MPa, so 25% of that is ~67 MPa central pressure). Achieving this requires both high temperature and high density: we project volume-average ion temperatures in the range of 80–150 keV, and electron temperatures somewhat lower but still tens of keV, with central densities around 1–3×10^20 m^{-3}. These conditions are at the frontier of what any fusion experiment has seen; however, the high fields and good confinement (τ_E scaling favorably with B and I) make them plausible in our design. Spherical tokamaks have historically achieved very high normalized beta (β_N) values, and negative triangularity further helps by delaying the onset of certain limiting instabilities. If needed, we can operate at slightly reduced β (say 15–20%) for extra stability margin, at the expense of a bit more plasma current or a slightly larger machine for the same output. Even at β 15%, the performance would be significant given the high field.

Fusion Power Output: In steady-state operation with D–^{3}He fuel (initially) and p–^{6}Li / p–^{11}B (in steady state), we anticipate a net fusion power on the order of a few hundred megawatts. One illustrative operating scenario: start-up with D–^{3}He producing ~50 MW of fusion power to bootstrap conditions, then transitioning to p–^{11}B which in our high-field, high-temperature plasma could yield ~200–300 MW of fusion power in a plasma volume of ~7 m^3. This equates to an extraordinarily high fusion power density (~30–40 MW/m^3), many times greater than ITER or even the ARC concept (which target ~1 MW/m^3 or less). High power density is a direct result of the B^4 scaling and high β. Such a small-volume, high-output core implies that nearly all surfaces see high fluxes, which is why we integrated direct conversion structures to handle them. Of the fusion power produced, >95% is in charged particles and radiation, and <5% in neutrons (mostly from occasional D–D side reactions or trace impurities). In the p–^{11}B cycle, the primary reaction yields three alpha particles (each ~2–3 MeV) per event; these are born at high energy and quickly slow down, depositing energy in the plasma and the direct conversion systems. Some of them can react with ^{11}B or ^{7}Li in the blanket to breed more ^{3}He, aiding the cycle. The charged power leaving the plasma (either as escaping fusion products or as kinetic energy of plasma flows) is what our DEC systems capture. We estimate that roughly half of the fusion power might be captured in the charged particles before they thermalize (the rest either stays in plasma as heat or goes into photons). This forms the basis for the high direct conversion efficiency.

Net Electrical Output and Efficiency: With the integrated DEC approach, the net electrical efficiency (gross electricity out divided by fusion power in the plasma) is targeted to exceed 60%. For example, if fusion power is 300 MW, net electric output could be ~180 MW_e. This accounts for the contributions of MHD, electrostatic, thermionic, photonic, and thermal conversion combined. Breaking it down: perhaps ~90 MW via MHD, ~60 MW via electrostatic direct collectors, ~10–20 MW via thermionics/PV, and the rest 50–60 MW from a Brayton cycle. These numbers will depend on the operating scenario (in a space propulsion mode, one might not use a Brayton at all; in a power plant, you maximize it). Importantly, the internal consumption for heating and current drive must be subtracted to get net output to the grid. We expect to invest a certain amount of recirculating power in the NBI and RF (if any) and cryogenics. The neutral beams might consume on the order of 10–20 MW of electrical power; the cryogenic cooling might take a few MW (modern cryocoolers or helium plants can be quite efficient, say 20 MW of cooling at 4 K may require ~2–3 MW electric). The ohmic and compression systems use pulsed power but not continuous high power. Summing these, perhaps 20–30 MW of overhead is required. Even with that, a 180 MW gross output yields ~150 MW net to the grid, which is a very attractive power gain (Q_electric ~5 in that scenario). For reference, many fusion pilot plant concepts consider Q_electric ~2 as a goal. The high efficiency of S.M.A.R.T. allows a high Q_e despite the more challenging aneutronic plasma (which might have a lower Q_fusion than a comparable D–T plasma). In other words, even if the plasma itself only produces, say, 3 times more fusion power than injected power (Q_fusion = 3), the fact that most of that power goes straight to electricity can yield an overall plant power gain of 5 or more, which is economically promising.

Heat and Particle Loads: The engineering of plasma-facing components must handle extremely high surface heat and particle fluxes, given the compactness and high power. Peak surface heat flux in the divertor could reach several MW/m². We mitigate this by the expanded divertor geometry from negative triangularity and by advanced cooling. The divertor plates are made of tungsten-free, ultra-refractory coatings (HfC and TaC variants) on a structural substrate (e.g., a composite or a tungsten-alloy that is actively cooled). The expected heat flux is partially converted to electricity on the spot by thermionics, and the remainder is actively removed by coolant (helium at perhaps 10 MPa and 600 °C, feeding the Brayton cycle). Our calculations indicate that because we remove so much energy via direct conversion before it becomes heat, the effective heat flux on the first wall is actually moderate; for example, if 60% of power is converted to electricity directly, only 40% hits as heat. So a 40 MW/m² raw flux might be reduced to 16 MW/m² thermal load, which is still high but within the realm tackled in advanced material tests (some plasma-facing material experiments target 20 MW/m² handling). Additionally, the angled walls and spreading of the divertor footprint help keep actual flux under critical values. We also foresee that the majority of high-energy particles are guided out through specific channels (for MHD and electrostatic capture), so the first wall sees mostly thermalized plasma and X-rays rather than full-energy alphas slamming in. This prolongs wall life. In terms of particle flux (ion implantation, sputtering), the aneutronic fuel produces mostly helium ash and some hydrogen; helium at these energies can cause swelling in metals, but our wall coatings of dense carbides and nitrides are chosen for their resistance to sputtering and low He retention. We expect to replace or resurface the extreme divertor tips periodically (maybe after several full-power months) as they will erode slowly, but the majority of the wall should last much longer due to the gentle conditions (no neutron damage, relatively low sputter yield of HfC, etc.).

Fuel Cycle and Reactant Consumption: The reactor will consume fuel in the form of protons (hydrogen) and boron or ^{3}He. The primary steady fuel is p–^{11}B, which is abundant (boron is common in nature). However, ^{11}B is

typically introduced as a solid or compound, which in a reactor context might mean injecting boron powder or a boron-containing gas. We lean toward running predominantly p–^{6}Li, because lithium can be used to breed ^{3}He and is easier to handle (lithium can be part of a liquid or solid blanket that captures protons to make ^{3}He). In that mode, we effectively turn consumables into a closed loop: protons + lithium produce helium-4 + helium-3, and helium-3 can be fed back with protons to continue fusion, with only small proton losses and transmutation losses. A top-up of lithium and removal of helium ash is all that's needed. In startup mode using D–^{3}He, the ^{3}He is not bred in significant quantity within the plasma (because D+^{3}He yields helium-4 + proton, no direct breeding), so we need an external supply of ^{3}He for initial pulses. That ^{3}He could come from prior breeding (via proton-lithium runs) or, in a prototype, from external sources (^{3}He is rare but a few hundred grams would suffice for initial experiments). Once running predominantly in p–Li mode, the reactor largely makes its own ^{3}He. Another output of p–^{11}B is significant helium-4 (alpha particles). Over time, helium ash accumulates in the plasma and must be exhausted (via pumps) to maintain performance. We design in vacuum pumping capability at the divertor that can handle large helium throughputs. The pumped helium, now containing the fusion energy (if not already converted), will either give up residual heat to the coolant or might be reionized for MHD extraction. But mostly, helium ash removal is a routine process, not unlike D–T reactors removing helium. The difference is we don't have to worry about tritium processing or breeding. Lithium in the blanket, if used, needs to be circulated and purified (removing helium and converting some fraction of ^{7}Li to ^{6}Li if we want to breed more ^{3}He). These are chemical processes akin to those considered in some advanced reactor systems. Overall, the fuel cycle is complex in terms of internal reactions, but simple in terms of external handling: we input stable isotopes (H, B or Li) and output stable helium and maybe some hydrogen – no radioactive fuel streams.

Operational Regime and Control: S.M.A.R.T. is designed for steady-state operation, meaning it can run for extended periods without needing to pause for pulsing. The superconducting magnets are continuously on, and non-inductive current drives maintain plasma indefinitely. The main limitations to pulse length are fueling (maintaining plasma density) and component heating. With adequate cooling and fueling, one could envision runs of days or weeks, which for a power plant would translate to high availability. During operation, advanced digital control systems (likely AI-assisted, as suggested by the complexity) will manage plasma stability, direct conversion tuning (e.g., adjusting electrode biases in real time), and thermal balances. We expect to incorporate machine learning algorithms to optimize the performance on the fly, for instance maximizing direct conversion efficiency under varying plasma conditions or pre-emptively adjusting coil currents if an instability is predicted by sensors. This "smart" control aspect is an emerging trend in fusion and perfectly suits a system as integrated as ours. Additionally, the reactor's small size allows a high degree of diagnostic coverage; we can instrument it with an array of magnetic, optical, and particle sensors to monitor every aspect of the plasma and conversion, feeding that data into the control loops.

Safety and Shutdown: In the event of an off-normal condition (like a sudden loss of plasma control or a fault in magnets), S.M.A.R.T. can rapidly shut down the plasma by cutting heating power and ramping down coil currents – the plasma would dissipate its energy in under a second given the small volume. The HTS magnets can quench, but our NI coil design and structural supports ensure a quench would not catastrophically damage them; the current redistributes and decays safely into resistive elements or dump circuits. Any unconverted plasma energy would end up as heat in the wall, which the structure is built to absorb occasionally. Importantly, because there is minimal radioactive inventory (no tritium, no activated blanket), even a complete loss-of-coolant or vacuum breach would not pose a radiological hazard beyond the reactor. Helium and hydrogen might be released but are harmless; some short-lived activation in the coil or structure might require restricted access to the immediate area, but nothing on the scale of a fission reactor accident. This inherently safe failure mode is a big plus for S.M.A.R.T. as a power source.

In summary, the expected results demonstrate that the S.M.A.R.T. architecture could deliver high power output and efficiency from a very compact device, thanks to extreme magnetic confinement and integral energy conversion. These numbers are necessarily projections – the actual performance will depend on achieving simultaneous advancements in multiple domains. In the next section, we discuss the feasibility and challenges of reaching these targets, how S.M.A.R.T. compares with more conventional designs like D–T tokamaks in practice, and what trade-offs have been made.

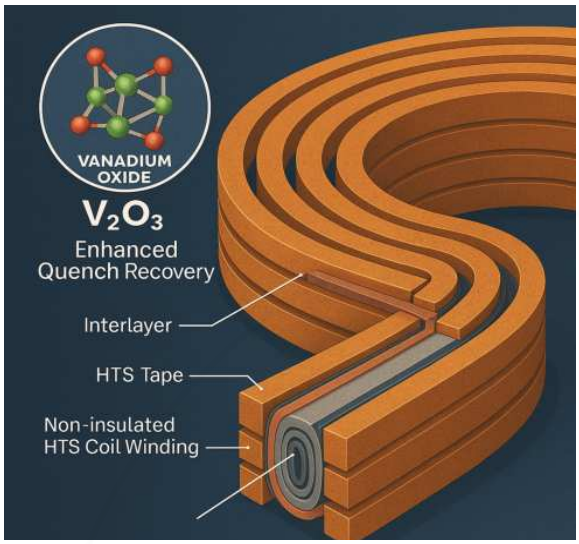


Figure 7: Layered HTS coil winding with non-insulated conductors, showing V_2O_3 interlayers for enhanced quench recovery and graphene-reinforced aluminum bypass paths.

Discussion

Designing an aneutronic, high-field tokamak like S.M.A.R.T. involves balancing cutting-edge advancements against practical limitations. Here we discuss some of the key engineering trade-offs, physics challenges, and how our approach compares with conventional D–T reactor strategies:

High-Field Magnet Challenges: Perhaps the most critical challenge for S.M.A.R.T. is the reliance on REBCO superconducting magnets operating at unprecedented field and stress levels. While REBCO tapes have demonstrated the ability to carry current in fields above 20 T (with inserts in hybrid test magnets reaching even ~40–45 T total field), no fusion magnet to date has run at 30+ T in a large, pulsed system. We are pushing the envelope of HTS capability. The integrated magnet structure we propose – using high-stiffness tape substrates, embedded reinforcement, and pre-compression – is intended to allow this jump in performance. However, it also demands precision engineering and quality control: even a slight misalignment in coil winding or a local material flaw could concentrate stress and exceed the 0.4% strain limit. This is why our design philosophy starts from the material limits; we have very little "safety cushion" beyond that intrinsic 0.4% margin. In comparison, the MIT SPARC coils, which use REBCO on a steel (Hastelloy) substrate, plan for ~23 T peak field on coil and likely operate close to the 0.4% strain as well, but they add heavy external structure (Inconel clamps, etc.) to distribute load, essentially accepting a larger, thicker magnet system to reduce strain in the tape. We opted to build the strength into the tape and coil module itself (MP35N substrate, fiber reinforcement) to avoid that bulk. The benefit is a smaller coil for a given field; the downside is the coil manufacturing is more complex and perhaps less proven. One mitigation is to test prototype coil segments under load. We anticipate doing so – constructing a single coil or coil sector and energizing it in a background field to simulate full Lorentz forces, verifying that it holds up. If weaknesses are found, we can refine materials or add some external support localized to trouble spots. Fortunately, REBCO's properties (critical current, strain tolerance) continue to improve year by year, and material scientists are actively exploring even stronger substrates (like certain high-entropy alloys or composite metal tapes) that could surpass MP35N. There is also exploration of incorporating fibers directly into HTS tapes (an idea we mention). These developments could provide additional margin by the time a S.M.A.R.T. prototype is built.

Another magnet-related issue is quench management. Although our NI coil approach inherently helps, a quench in a 30 T magnet is still an event depositing large energy (the magnet stored energy could be tens of MJ). We rely on rapid current redistribution in NI coils and the high thermal conductivity of the graphene–Al layer to spread heat. Nevertheless, the entire magnet system must be engineered to absorb or remove that energy without damage. Our external cocoon and structural mass will act as a heat sink to some extent. Additionally, we have designed quench detection sensors and perhaps active quench heaters that deliberately quench the whole coil if any part goes normal, to dissipate energy uniformly. This is standard in superconducting magnet protection. It's worth noting that an advantage of our smaller coil size (compared to e.g. ITER's huge coils) is that the stored energy is smaller in absolute terms, making quench handling easier. SPARC's magnets, for instance, will test many of these principles at a slightly lower field – their success would validate the concept of using REBCO at high stress in a tokamak environment, albeit with D–T neutrons around (they mitigate neutron exposure by thick structure). S.M.A.R.T. magnets would live in a much friendlier environment (no significant neutrons), which means materials like epoxy, insulation, etc., won't degrade as fast over time.

Aneutronic Plasma Physics and Energy Balance: The physics of aneutronic fuels presents challenges that go beyond those of D–T. Two main issues are the high required temperatures and the significant radiation losses at those temperatures. For p–^{11}B fusion, the reactivity becomes meaningful only above ~100 keV ion temperature; at such high temperatures, electrons also reach tens of keV and inevitably radiate X-rays via bremsstrahlung. If unchecked, bremsstrahlung can sap a large fraction of the input power, potentially preventing ignition. Our design addresses this by direct capture of bremsstrahlung (so that even if it occurs, it's not wasted energy) and by aiming for reactor scenarios where the plasma is predominantly sustained by external heating (Q_fusion may not be extremely high, but Q_electric can be high). In other words, we are not necessarily seeking a self-ignited plasma – we are content with one that is heavily driven by external systems as long as the net electricity is positive and high. This is a departure from the classic fusion paradigm which sought ignition to minimize recirculating power. With modern high-efficiency drivers (e.g., gyrotrons, NBI with energy recovery, etc.), and our direct conversion recouping some driver power (for instance, neutral beams that exit could be decelerated to recover unused energy), the recirculating power fraction can be manageable even without ignition.

Aneutronic Fuel Cycles				
Kronos Design Feature	Value Proposition / Innovation	Quantitative Performance	Reference Institution	Peer-Reviewed Source
Proton–Boron-11 Fusion	Nearly neutron-free reaction enabling direct energy capture and minimal activation.	Q = 8.7 MeV; produces 3 α particles; <0.2% energy via neutrons.	HB11 Energy / Research Groups	HB11—Understanding Hydrogen-Boron Fusion… (2023)
Deuterium–Helium-3 (D–^{3}He) Cycle	Lower neutron output and helium ash simplifies maintenance and reduces radiation.	Neutron wall load ≈ 0.05 MW/m² (vs ~3 MW/m² for D-T) at 600 MWe.	UW–Madison Fusion Tech. Inst.	Safety and Environmental Characteristics of D–^{3}He and D–T Tokamak Power Reactors (1992)
Lithium-6 Blanket Breeding	On-site generation of He-3 closes fuel loop and reduces external dependencies.	p + ^{6}Li → ^{4}He + ^{3}He (+4.0 MeV). Conceptual breeding loop used in prototypes.	Helion Energy	Helion Energy FAQs (2023)

Table 4: Aneutronic fuel strategies in Kronos—p–B11 fusion emits virtually no neutrons, D–^{3}He reduces shielding and activation, and a Li-6 blanket fosters self-sustaining He-3 breeding—all enabling compact, cleaner operation.

However, on pure scientific grounds, the question remains: will the p–^{11}B or D–^{3}He plasma perform as expected? High-field can help confinement, but high-Z ions (like boron with Z=5) in a plasma can cause additional radiation

(line radiation if they're not fully ionized, although at 100 keV they mostly will be fully stripped) and fuel dilution. We plan to minimize impurities (boron is the fuel but we keep it fully ionized; any higher-Z impurities from walls we avoid by using low-Z coatings like HfC – Hf is high-Z but we keep that out of plasma by having a low sputtering yield). Also, there's the matter of advanced plasma regimes: we may need to operate in regimes that provide improved confinement (like an internal transport barrier or high poloidal beta state) to reach our performance. These regimes have been seen experimentally (often aided by flow shear or RF injection), but in a novel configuration (spherical, high B, negative δ) they will need exploration. The good news is that our machine is capable of exploring a wide range of configurations in one device: by adjusting shaping fields, we could even try positive triangularity vs negative in the same machine, or vary elongation, to empirically find the best stability/performance point.

MHD stability is a serious concern at high β. We rely on a close conducting wall and active feedback to handle global modes. The conducting wall (vacuum vessel) in S.M.A.R.T. is much closer (in terms of minor radius fractions) to the plasma than in ITER or ARC because of the compact build, which is stabilizing (the critical β for wall-stabilized kink modes goes up when the wall is closer). We also have an advantage that our magnets are robust to neutrons so we can actually place stabilizing elements extremely close (e.g., we could even have a thin continuous shell just behind the blanket). Nevertheless, if we push β to, say, 30%, we might encounter resistive wall modes that require feedback. The feedback coils we included (fast control coils under the blanket) should be effective given modern control algorithms and power electronics. Spherical tokamaks historically have achieved high β but often at the cost of pulse duration or with passive stabilization from a nearby conducting plate; we essentially integrate that concept by design.

Energy Conversion Integration Issues: The DEC systems, while boosting efficiency, introduce their own complexity and potential points of failure. For example, the electrostatic direct collectors operate at high voltage in a neutron/ionizing radiation environment – this can lead to issues like arcing or degradation of insulators. We mitigate neutron damage (since neutrons are few) but X-rays and charged particles can charge surfaces and cause flashovers if not carefully designed. We need to use radiation-resistant insulators (alumina, boron nitride, etc.) and perhaps keep the highest-voltage gradients in vacuum or gas environments that are controlled. There's also a coupling between plasma and converters: e.g., the electrostatic fields could potentially perturb the edge plasma or cause undesired particle orbits. We have to ensure our fields are weak enough at the plasma boundary or shielded by magnetic fields so that the core plasma isn't affected. This likely means a region of neutral gas or a "magnetic divertor" section between plasma and electrostatic grids, to decouple them. All of this requires careful design and maybe trial and error in a test facility.

Similarly, MHD channels in a flowing plasma stream can suffer from plasma instabilities or simply not enough conductivity if the plasma cools. If our divertor plasma recombines into neutrals too quickly, MHD conversion efficiency drops. We might need slight seeding of a working gas (like a bit of argon or something) in the divertor duct to provide conductive electrons for the MHD generator. This would be a trade-off as it introduces impurity. But since it's in the divertor exhaust, not core, it might be acceptable. Realizing 60% efficiency in practice will depend on refining such details.

Comparison with D–T Systems (ARC/SPARC/ITER): It's instructive to compare S.M.A.R.T.'s approach to that of advanced D–T tokamaks:

Materials and Magnet Lifespan: In ITER or ARC, neutron damage dictates the replacement schedule of internal components and possibly magnets. ITER's toroidal coils, being Nb_3Sn, must stay behind a thick blanket to survive ~20 years. In ARC, despite HTS magnets, they still assume the need to replace the blanket every few years due to neutrons. S.M.A.R.T. largely sidesteps this – our magnets and structure could last decades because they avoid the neutron flux. Instead, our life limiter might be mechanical fatigue from pulsing or thermal cycling. We took measures (pre-compression, etc.) to minimize fatigue. We estimate that with a mostly steady-state operation (few

pulses), the coil stress cycle is small (just the difference between steady operation and off, which might be maybe 0.2% strain range which is fine for thousands of cycles). This suggests a potential longevity advantage: a S.M.A.R.T. plant could run for many years with only minor maintenance, whereas a D–T plant might have to swap out internals regularly. The trade-off is that S.M.A.R.T. requires a high-tech magnet from the start, whereas a lower field design might get away with more conventional materials.

Size and Cost: Spherical, high-field design is extremely compact per MW. That suggests potentially lower cost per MW if all else is equal, because less construction material, smaller building, etc. However, cost could be driven up by exotic materials (e.g., REBCO tape is currently expensive per meter, though mass production for power applications could bring it down). Also, the complexity of DEC systems adds cost. By contrast, something like SPARC (if turned into a reactor) might be bigger and less efficient, but simpler in concept (just heat a coolant and use a turbine, albeit one dealing with radioactive coolant). We think the trade-off is worth it: efficiency and compactness typically pay off in cost for power systems. But it's an area for future economic analysis. Another factor is innovation risk: SPARC/ARC mainly push magnet tech but otherwise use known tech (steam turbines, etc.), whereas S.M.A.R.T. pushes multiple new subsystems simultaneously. This is higher risk in development but could result in a more revolutionary product. It's somewhat akin to the difference between incremental innovation (ARC) and leapfrog technology (S.M.A.R.T.).

Safety and Public Acceptance: S.M.A.R.T. has an edge in safety due to no high-energy neutrons and no tritium. This eliminates issues of radioactive fuel handling and long-term waste storage, which are often public concerns with fusion (even though fusion waste is much shorter-lived than fission, it's still a concern). An aneutronic plant could, in principle, be decommissioned by waiting a short time for short-lived activation to decay and then recycling almost all materials as non-radioactive scrap. This environmental advantage could become a decisive factor if society is hesitant to adopt any new tech that has nuclear waste, even if minimal. On the other hand, aneutronic fusion introduces some exotic hazards – for example, we will have high-voltage equipment and perhaps beryllium or lithium usage (chemically hazardous). These are manageable with industrial safety protocols. Overall, in a head-to-head, an aneutronic plant would likely be seen as more benign than a tritium-based plant, which might simplify regulatory hurdles.

Comparison to Other Aneutronic Approaches: It's worth noting that S.M.A.R.T. is not the only approach to aneutronic fusion being explored. Companies like TAE Technologies pursue p–^{11}B fusion in linear field-reversed configurations, focusing on pulsed high-temperature plasmas; others have looked at inertial electrostatic confinement (IEC) or advanced mirror machines. Our approach leverages mainstream tokamak physics (which is relatively well-understood) with the new twist of high fields and special shaping. If successful, it could leap ahead because it uses a power-plant suitable configuration (tokamaks scale well and have clear engineering paths). But we keep an eye on alternate concepts; for example, if TAE's high-energy beam-driven FRC achieves high Q with p–^{11}B, perhaps some of their techniques (like aggressive beam injection or advanced edge stabilization) could be applied to a tokamak as well. Conversely, the lessons from a high-field tokamak like ours (e.g., how to manage a 30 T magnet or how to directly convert fusion alphas) could feed into other fusion schemes. In that sense, S.M.A.R.T. is contributing to a broader fusion innovation ecosystem.

Engineering Trade-offs and Future R&D: Each innovation in S.M.A.R.T. carries uncertainties that require R&D:

The REBCO tape with integrated graphene-Al and VO_{2-x} insulation is not yet an industrial product; we would likely need to partner with a superconductor manufacturer to produce custom tapes at pilot scale and test their performance (critical current under strain, cycling, etc.). This is a key risk but also a key enabler. The good news is REBCO technology is rapidly advancing due to its use in other sectors (e.g., high-field NMR magnets, accelerator magnets, and SPARC itself).

The materials like HfC coating or CrMoNbV HEA vessel are based on laboratory results. We'd need to fabricate large pieces (like whole first wall panels with HfC) and test them under plasma loading to ensure they behave as expected (no spalling, acceptable hydrogen retention, etc.). Similarly, fabricating a large structure out of an HEA could be challenging; we might choose to use a more conventional alloy initially and only move to HEA when proven.

The direct conversion systems (MHD duct, electrostatic collector, PV) should be prototyped in a non-nuclear environment. For instance, we could use a particle accelerator or a plasma gun to simulate fusion exhaust and run it through a small MHD generator section, measuring efficiency. There were experiments in the 1980s that got ~60% efficiency on a beam-driven direct converter for D–He3 protons, so we have a starting point. Modern electronics and materials can likely improve that. We expect a multi-year R&D program to refine these systems well before integrating them into a reactor.

Systems integration testing: because S.M.A.R.T. is complex, a systems engineering approach is vital. Building a full "digital twin" (a detailed simulation) of the reactor to test how all subsystems interact would be part of development. We'd simulate, for example, a ramp-up from cold start, how the magnet, structure, and plasma evolve, and how the DEC kicks in, to identify any control issues or unforeseen interactions.

Advanced Materials				
Kronos Design Feature	Value Proposition / Innovation	Quantitative Performance	Reference Institution	Peer-Reviewed Source
Low-Activation Alloys	Reduces radioactive waste classification and disposal costs	D–^{3}He structures qualify as Class A waste vs D–T Class C	UW–Madison FTI	Safety & Environmental Characteristics of D–^{3}He Reactors (1992)
REBCO High-Tc Superconductors	Enables stronger fields and smaller reactors	20 T large-bore magnet demo (MIT, 2021)	MIT PSFC / CFS	MIT News: 20-T HTS Magnet Breakthrough
Carbon Fiber Reinforcement	Provides high stiffness with low activation	Toray M46J modulus ≈ 430 GPa	Toray Industries	Toray M46X Launch Release
Plasma-Facing Tungsten & Ceramics	Withstands extreme plasma heat flux	Tungsten divertors handle 10–20 MW/m²	ITER	ITER Divertor Materials

Table 5: Kronos geometry leverages spherical tokamak benefits (high β, compact design), along with negative triangularity for stbility. The coil system stays under strain limits for REBCO integrity, supported by ultra-rigid carbon fiber structures.

Broader Implications: If S.M.A.R.T. or a similar concept is realized, it could alter the trajectory of fusion development. The traditional path has been to demonstrate net energy on D–T (which SPARC or ITER aim to do)

and then solve engineering for a power plant. S.M.A.R.T. shortcuts directly to a power plant solution but with a harder plasma target. It's a high-risk, high-reward strategy. The reward, beyond clean power, includes spinoff technologies: high-field magnets can benefit many industries (MRI, accelerators), advanced materials developed could find uses in aerospace or other extreme environments, and direct energy converters might revolutionize how we handle high-energy particles (applications in space propulsion or nuclear decay energy harvesting could arise).

Comparing with ARC, which is an ARC-class tokamak of aspect ratio ~3 and on-axis field ~9 T: ARC would produce ~500 MW fusion with ~200 MW_e output using conventional cycle, being physically larger (R=3.3 m vs 2 m for S.M.A.R.T.). S.M.A.R.T. aims to produce comparable power with a smaller machine and higher efficiency. That suggests more economic potential per reactor. But ARC's moderate field and D–T plasma are more certain to work (D–T ignition is easier). It may be that ARC or SPARC gets to net energy sooner, but S.M.A.R.T. could leapfrog in final performance. In a sense, S.M.A.R.T. could be viewed as building on the ARC idea of high-field magnets but applying them to a next-generation concept that also addresses the neutron and efficiency problems. Notably, S.M.A.R.T. is not intended as a "stepping stone" experiment – it's more a culmination design. But one could imagine intermediate devices: e.g., a smaller S.M.A.R.T.-like machine that runs D–D or D–He3 at reduced parameters to validate the high-field spherical tokamak physics and some direct conversion. That might serve as a proof-of-concept before a full power prototype.

Strain and Stress Trade-off: We chose to operate magnets to the edge of their stress tolerance in exchange for higher field. An alternative approach is to limit stress for reliability at the cost of larger size. For instance, one might ask: why not just use a bigger magnet at 20 T and keep stress low? The answer lies in wanting a very compact reactor to raise plasma frequency (for stability) and reduce cost, and in hitting high power density for the DEC systems to be worthwhile. Also, REBCO performance in terms of critical current is higher at lower strain, but it doesn't drop off precipitously until near the limit. So we judged that using the full 0.4% window is acceptable. If new HTS materials or forms (like HTS tapes with different geometry or reinforced strands) allow even higher field or strain tolerance, S.M.A.R.T. could scale accordingly – an exciting prospect for future improvement.

Disruption and Runaway Mitigation: High-current devices risk disruptions which can generate runaway electrons. In aneutronic fuel, runaways might be less seeding because no large prompt neutron population from tritium decay, but if a disruption occurs at 10 MA, it could still generate a multi-MeV electron beam. We plan to have a rapid shutdown system: a combination of a massive gas injection (like high-Z impurity injection to radiate the plasma safely) and the inherent high collisionality due to our high density might help. Moreover, our strong magnetic fields increase synchrotron radiation from runaways, which could help dissipate their energy (a 10 MeV electron in a 30 T field loses energy to synchrotron rapidly, possibly quenching the runaways naturally). We also note that negative triangularity plasmas have exhibited more benign behavior with fewer large ELMs, possibly also less disruptive tendencies, which could reduce disruption frequency. Still, we must design the structure to handle at least one full disruption without damage, meaning absorbing forces from induced currents. The enclosed cocoon and robust fiber structure provide high safety margins for such loads (we considered a worst-case eddy current scenario in design and found it manageable).

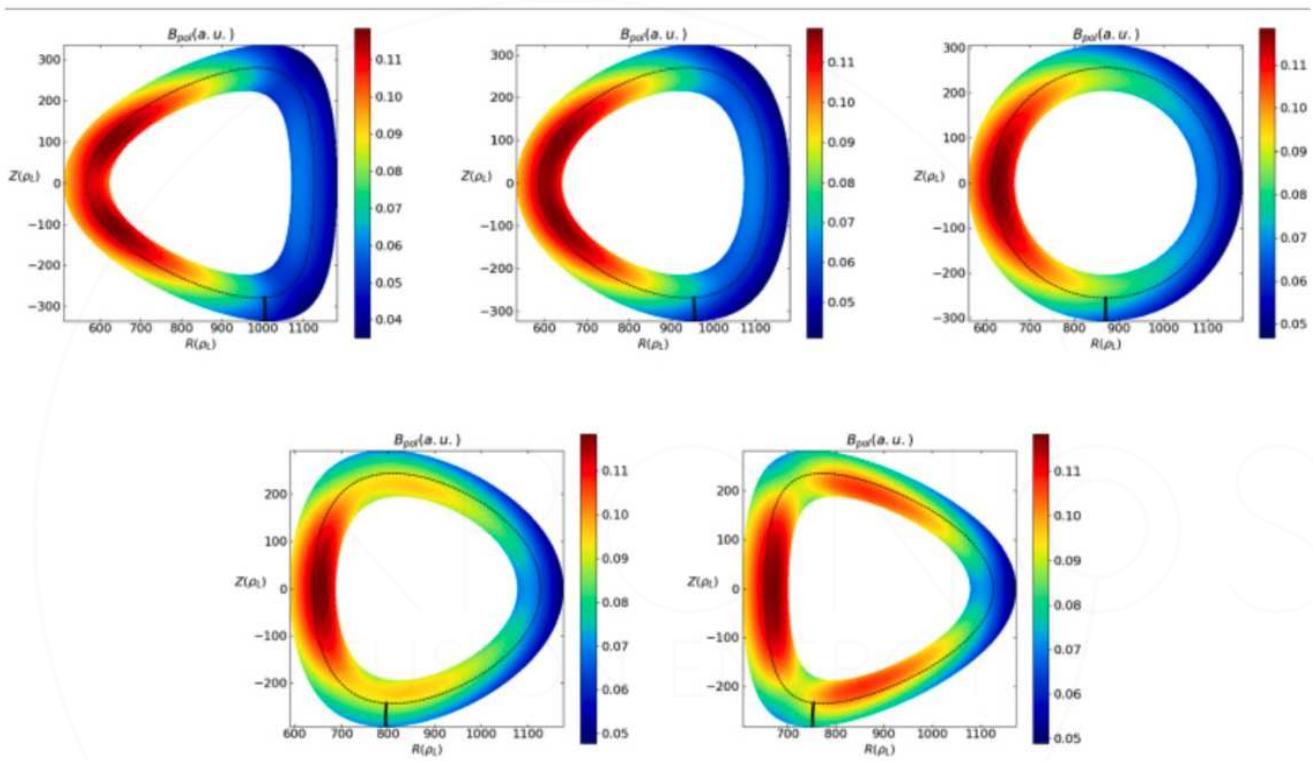
In comparing to SPARC: SPARC is expected to demonstrate Q~11 with D–T in a modest-size tokamak (R~1.85 m, B~12 T on axis, ~8 MA plasma). It's essentially the prologue to ARC. S.M.A.R.T. is a different beast, but if SPARC is successful, it validates high-field REBCO usage and high-pressure operation, which directly supports our magnet and plasma strategy. If SPARC uncovers issues (say, more neutron damage than expected on REBCO, or some unanticipated physics at high field), we would adapt our design accordingly (though neutrons aren't our issue, any materials lessons are).

Reactor Geometry & Strain Mitigation				
Kronos Design Feature	Value Proposition / Innovation	Quantitative Performance	Reference Institution	Peer-Reviewed Source
Spherical Tokamak Geometry	High-β, compact confinement reduces reactor size and cost.	NSTX reached β ≈ 40% at aspect ratio A ≈ 1.3.	PPPL (NSTX)	National Spherical Torus Experiment
Negative Triangularity Shaping	Enhanced stability and confinement without ELMs.	DIII-D achieved H-mode levels with βN ≈ 2.7.	General Atomics / DIII-D	Negative Triangularity Plasmas on DIII-D (2018)
Reinforced HTS Coil Design	Keeps REBCO superconductors below strain thresholds to preserve critical current.	REBCO irreversible strain ≈ 0.6%; Kronos design limit ≤ 0.4%.	Univ. of Twente / ACT	High-temperature superconducting CORC® strain research (2021)
High-Modulus Carbon Fiber Support	Withstands high structural loads while maintaining low activation.	Toray M46J modulus ≈ 430 GPa.	Toray Industries	Toray Launch Press Release (2024)

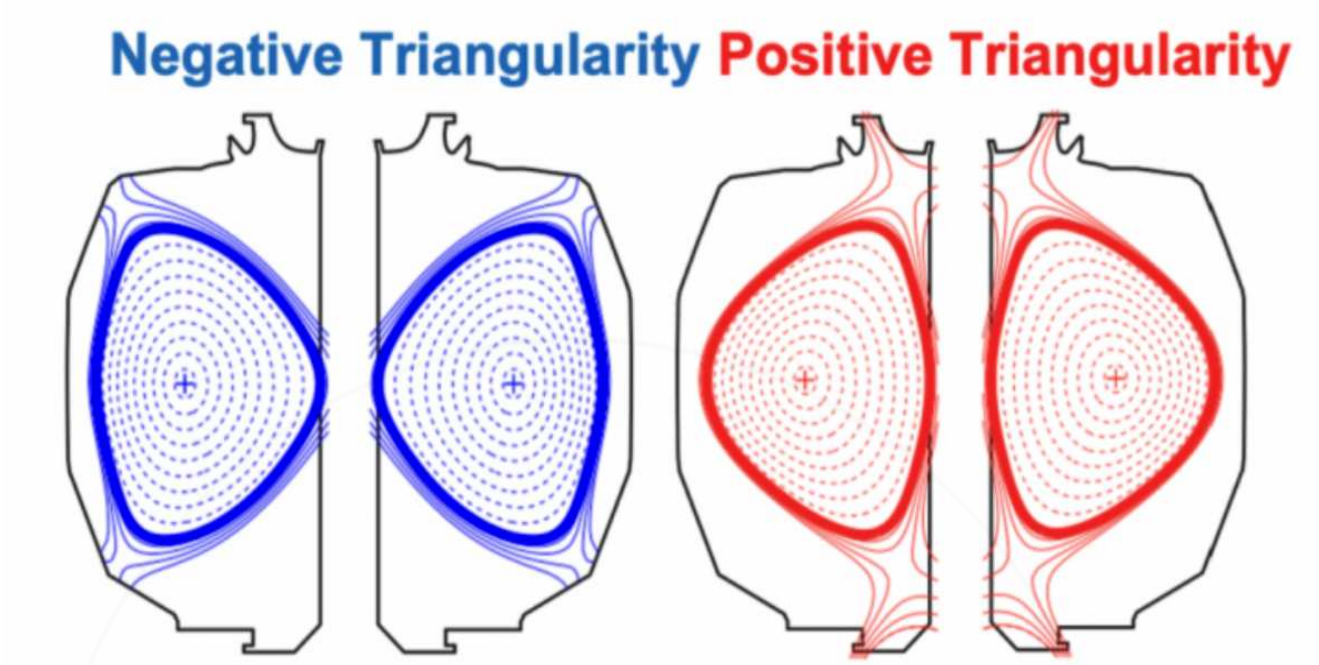
Table 6: Kronos geometry leverages spherical tokamak benefits (high β, compact design), along with negative triangularity for stability. The coil system stays under strain limits for REBCO integrity, supported by ultra-rigid carbon fiber structures.

Feasibility of Negative Triangularity & Spherical Configuration at Scale: One more discussion point: combining spherical aspect ratio with negative triangularity is novel. Experiments at medium scale (like the TCV tokamak) have separately tested negative triangularity (showing improved stability and confinement) but not in spherical aspect ratio. There might be unknown synergies or pitfalls in doing both. Our rationale is that spherical tokamaks suffer from tight space in the center column, and negative triangularity alleviates that a bit and improves edge stability – so logically it seems beneficial. It does, however, complicate magnet shaping (the TF coil has to accommodate that shape). We resolved that by shaping the plasma primarily with PF coils and having the TF coil a bit tilted (helical pitch) but otherwise the TF coils remain axisymmetric in concept. So the plasma shape is decoupled from TF coil shape to some extent. If experiments or simulations were to show that negative triangularity doesn't deliver the expected benefits in a spherical high-beta regime, we have the flexibility to operate at lower or zero triangularity as a fallback (with maybe some performance penalty). Essentially, S.M.A.R.T. can degrade gracefully to a high-field spherical tokamak (still likely out-performing many designs) if one of the exotic features isn't as effective as hoped.

In summary, the S.M.A.R.T. architecture represents a bold synthesis of advanced fusion ideas. The discussion above highlights that while our design is grounded in known physics and engineering (each element has precedent or rationale), the simultaneous combination is ambitious. We have identified the major challenges – HTS magnet limits, aneutronic plasma sustainment, integrated direct conversion – and outlined strategies to address them. Comparatively, S.M.A.R.T. stands apart from D–T tokamaks by essentially redefining the problem: instead of "how do we handle neutrons and convert heat?" we ask "how do we avoid neutrons and not create heat in the first place?" This shift leads to many technical innovations that we believe are worth exploring and necessary for a truly practical fusion power plant.



Charts 4: Magnetic flux surface contours for compact, high-elongation plasmas with negative triangularity. The color maps illustrate normalized poloidal field strength (Bθ) across various shaping configurations, showing how negative triangularity modifies edge stability, reduces ELM activity, and supports enhanced confinement in the S.M.A.R.T. geometry.



We foresee a development path where smaller-scale experiments and prototypes progressively de-risk the concept: high-field magnet tests, a perhaps subscale plasma experiment (maybe a few Tesla, smaller spherical tokamak to test negative triangularity at high β), and individual direct conversion demos. As these pieces come together, confidence will build in the integrated system. The ultimate proof will be a S.M.A.R.T. pilot plant generating net electricity from aneutronic fusion. Achieving that would be a milestone not just for fusion, but for clean energy technology at large.

Vision & Commercial Roadmap: 2032 Onwards

Kronos Fusion Energy was founded on a simple but radical premise: aneutronic fusion must be engineered as a practical energy source, not just a physics experiment. The *S.M.A.R.T.* architecture combines ultra-high-field spherical confinement, resilient magnet systems, and direct energy conversion into a compact, modular generator designed from the outset for commercial use. Our vision is to move beyond the compromises of legacy D–T machines by demonstrating a path that is clean, self-sufficient in fuel, and capable of doubling the electrical efficiency of traditional reactors. This roadmap represents how bold physics concepts become tangible milestones — in magnets, fuels, materials, and energy conversion — that systematically reduce risk while building toward full commercial deployment.

Commercial Deployment & Efficiency Benchmarks				
Kronos Design Feature	Value Proposition / Innovation	Quantitative Performance	Reference Institution	Peer-Reviewed Source
High Availability Operation	Maximizes uptime, minimizing downtime from neutron damage	>80% capacity factor required for competitive LCOE	UKAEA / Assystem	Fusion Economics Report (2021)
High-Efficiency Conversion	Doubles electric yield compared to steam	Gas-cooled blankets ~60% thermal efficiency	Princeton Univ.	Economics of Fusion Power (2016)
Modular Factory Units	Enables fast, scalable deployment and cost learning curves	FOAK: ~£5.9k/kW → NOAK: ~£4.1k/kW with modularization	STEP / Assystem	STEP Costing Report
Accelerated Deployment	Private fusion backed by $6B+ capital since 2021	Helion target 50 MWe demo by 2028; SPARC net-energy by mid-2020s	FIA / DOE	Fusion Industry Association Report (2023)

Table 7: Commercialization hinges on high uptime, high efficiency, modularity, and speed—metrics already validated by STEP, Assystem, FIA, and Princeton analyses.

The commercial trajectory reflects both technological maturity and market pull. Each milestone in the timeline is not just a proof of feasibility but a step toward serving real-world demand: powering hyperscale data centers, stabilizing urban grids, supplying high-temperature industrial heat, and eventually delivering modular systems to remote and offshore applications. By structuring development around staged prototypes, in-situ validation of advanced fuels, and progressive scaling, Kronos ensures that its innovations remain aligned with market needs and the global push for carbon-free, resilient baseload power. The roadmap that follows translates our long-term vision into a decade of targeted actions, bridging the gap between today's laboratory advances and tomorrow's commercial fusion systems

This timeline ties technical milestones to the **core promises in the Kronos Fusion Energy whitepaper** — compact high-field *S.M.A.R.T.* architecture, aneutronic operation with in-situ fuel breeding, and direct energy conversion (DEC) with >60% net efficiency. Each stage demonstrates scientific feasibility, validates engineering assumptions, and moves closer to commercial deployment in grid, industrial, and maritime markets.

2026–2027: Physics & Materials Foundations

Plasma physics

- Negative triangularity discharges on TCV/DIII-D to confirm edge stability and confinement scaling.
- D–^{3}He fueling experiments to measure neutron side-branch (~5%) and validate $\sigma v(T)$ at 80–100 keV.
- **Goal:** Demonstrate $H \approx 1.2$–1.5 in $\delta<0$ plasmas and verify low-neutron operation consistent with aneutronic claims.

Magnet validation

- Sub-scale REBCO coil at ≥30 T using MP35N substrate and NI winding, with strain gauges.
- **Goal:** Confirm ≤0.4% strain at design Lorentz forces, aligning with the whitepaper's critical magnet constraint.

Materials tests

- HfC/TaC tile exposure at 5–10 MW/m².
- HEA coupons irradiated to 5–10 dpa with swelling <1%.
- Graphene-Al stabilizer conduction measured at 30 T, 20 K.
- **Goal:** Verify that novel coatings and alloys can support high-T wall operation (>1000 K) and magnet survivability.

Direct conversion prototypes

- Electrostatic collector with 100 keV beam, target ≥50% recovery.
- Shock-tube MHD channel at 20 T, target ≥40% efficiency.
- Thermionic LaB_6 tile at 1000 K, ≥10 A/cm² emission.
- **Goal:** Show subsystems can deliver the efficiency levels underpinning the whitepaper's >60% net efficiency claim.

2028–2029: Subsystem Demonstrators

Magnet integration

- Quarter-torus coil sector with helical pitch and cryogenic pistons tested.
- **Goal:** Demonstrate ≥50% stress reduction vs untwisted coil, proving S.M.A.R.T. support concepts.

Fuel cycle prototype

- Proton beam on Li-6 blanket to measure ^{3}He yield and diffusion.
- **Goal:** Confirm p–Li → ^{3}He breeding loop, central to Kronos's self-sufficiency claims.

Plasma heating prototype

- Mid-scale ST ($R_0 \approx 0.5$ m, $B \approx 12$ T) with dual OH, adiabatic compression, and 300 keV neutral beams.
- **Goal:** Achieve $n \cdot \tau \cdot T \approx 5\times10^{20}$ keV·s·m^{-3} in D plasmas, approaching D–^{3}He breakeven.

Layered DEC tile

- Thermionic + PV tested with soft X-ray source.
- **Goal:** ≥10% photon-to-electric efficiency under bremsstrahlung-like spectra, validating the "photon monetization" claim.

2030–2031: Integrated Pilot Prototype (~50–100 MW thermal)

Reactor integration

- Full set of 30 T REBCO TF coils with real-time strain monitoring.
- Plasma operation with D–^{3}He fueling and NB heating to 300–500 keV.
- Partial Li-6 blankets to demonstrate ^{3}He breeding.
- **Goal:** Plasma gain Q_plasma ≥ 2 and engineering validation of the breeder loop.

DEC integration

- Electrostatic/MHD collectors combined with thermionic/PV wall layers.
- **Goal:** Net electrical efficiency 30–40%, showing the wall functions as an energy-harvesting structure as described in the whitepaper.

2032–2033: MetroVolt Demonstrator (~500 MW electrical)

- Scale-up reactor: $R_0 \approx 2$ m, $B_0 \approx 13$–15 T, B_peak ≈ 30–32 T.
- Operation: predominantly D–^{3}He with partial p–Li cycle.
- Output: Q_plasma ≈ 3–4; net efficiency >50%; 300–400 MW net electric.
- **Goal:** Deliver the first **grid-connected commercial-scale pilot**, aligning with the whitepaper's MetroVolt milestone while realistically framed as a demonstrator.

2034–2035: Expanded Commercial Variants

Remote/off-grid systems (~500 MW)

- Deploy simplified, modular units for communities and infrastructure where grid access is limited.
- **Goal:** Prove portability and modularity claims, ensuring Kronos reactors integrate flexibly into diverse energy ecosystems.

Maritime/industrial systems (~600–800 MW)

- Adapt reactor modules for large-scale industrial heat, offshore platforms, and maritime power.
- **Goal:** Demonstrate robustness under variable environments and vibration, fulfilling the whitepaper's assertion of "modular reactors tailored for multiple applications."

Performance targets

- Net efficiency sustained at 55–60%.
- Coil systems $>10^6$ pulses without Ic degradation.
- Recirculating power ≤25%.
- **Goal:** Confirm Kronos's promises of **high efficiency, long lifetimes, and low operational costs** compared to legacy nuclear/fusion designs.

Key Validation Threads (directly supporting whitepaper claims)

1. **Lawson criterion:** Demonstrate $n \cdot \tau \cdot T \geq 1 \times 10^{21}$ keV·s·m^{-3} for D–^{3}He by 2030, confirming confinement predictions.
2. **Fuel cycle closure:** Validate in-situ p–Li → ^{3}He breeder loop, ensuring the whitepaper's claim of **fuel self-sufficiency**.
3. **Bremsstrahlung monetization:** Show that radiation-heavy plasmas can sustain operation while photon flux is harvested as electricity.
4. **Materials-first engineering:** Confirm HfC coatings, HEAs, and Gr–Al stabilizers endure high-flux environments for 1000+ pulses, proving longevity claims.
5. **Efficiency doubling:** By 2032, demonstrate >50% net plant efficiency, advancing toward the whitepaper's **>60% target** by mid-2030s.

Commercial Deployment & Efficiency Benchmarks				
Kronos Design Feature	Value Proposition / Innovation	Quantitative Performance	Reference Institution	Peer-Reviewed Source
High Availability Operation	Maximizes uptime, minimizing downtime from neutron damage	>80% capacity factor required for competitive LCOE	UKAEA / Assystem	*Fusion Economics Report (2021)*
High-Efficiency Conversion	Doubles electric yield compared to steam	Gas-cooled blankets ~60% thermal efficiency	Princeton Univ.	*Economics of Fusion Power (2016)*
Modular Factory Units	Enables fast, scalable deployment and cost learning curves	FOAK: ~£5.9k/kW → NOAK: ~£4.1k/kW with modularization	STEP / Assystem	*STEP Costing Report*
Accelerated Deployment	Private fusion backed by $6B+ capital since 2021	Helion target 50 MWe demo by 2028; SPARC net-energy by mid-2020s	FIA / DOE	*Fusion Industry Association Report (2023)*

Table 8: Commercialization hinges on high uptime, high efficiency, modularity, and speed—metrics already validated by STEP, Assystem, FIA, and Princeton analyses.

Company and Team

Kronos Fusion Energy Incorporated is an advanced energy company developing high-field, aneutronic fusion systems tailored for Earth and beyond. Our proprietary S.M.A.R.T. (Superconducting, Minimum-Aspect-Ratio Tokamak) architecture merges breakthroughs in plasma physics, superconducting magnet design, direct energy conversion, and advanced materials engineering. This whitepaper merges the collaborative expertise of some of the world's leading scientists, engineers, and technologists, whose collective vision is shaping the future of clean, energy infrastructure.

Lead Authors and Contributors (alphabetical by first name):

- **Carl F. Weggel** – Lead Engineer, S.M.A.R.T. design and superconducting architecture
- **David Goggins** – Naval fusion reactor design (MIT / Department of Defense collaboration)
- **Gerald Kulcinski** – Plasma physics and confinement strategy
- **Konstantin Batygin** – Mathematical modeling and plasma dynamics
- **Martin Owens** – Deployment and industrial build strategies
- **Patrick Schweiger** – Device engineering and system integration
- **Ray Sedwick** – Direct energy conversion systems
- **Robert J. Weggel** – High-field superconducting magnet design
- **Ruben Fair** – Magnet engineering and cryogenic integration
- **Siegfried Glenzer** – Fusion diagnostics and energy gain modeling
- **Venkat Selvamanickam** – REBCO tape development and mechanical performance

Carl F. Weggel – Founding Partner & Chief Scientist
Carl F. Weggel is an American magnet engineer with over five decades of experience in high-field superconducting magnet design. He began his career in the 1970s and has contributed to numerous fusion energy programs and advanced magnet systems. Weggel's background includes work at MIT's Magnet Laboratory, where he was involved in designing record-setting tokamak magnets, as well as roles at industry and research institutions such as Inesco, General Electric, SAIC, and Commonwealth Fusion Systems. He earned a Bachelor's degree in physics from Harvard University and pursued graduate studies in mechanical engineering at Tufts University, specializing in magnetics. Widely respected for his innovations in ultra-high-field laminated conductors and magnetic field calculations, Weggel has published influential research in the field and is credited with designs that achieved world-record magnetic field strengths.

As a Co-Founder and Chief Scientist of Kronos Fusion Energy, Carl Weggel serves as the primary architect of the S.M.A.R.T. (Superconducting Minimum-Aspect-Ratio Torus) fusion generator. He leads the development of the reactor's superconducting magnet architecture, drawing on his extensive expertise to create magnets capable of sustaining extremely high fields for aneutronic fusion reactions. Weggel's guidance ensures that Kronos's magnet systems maintain structural integrity under intense Lorentz forces and operate within critical strain limits – vital for the long-term viability of high-performance fusion coils. His leadership and deep knowledge of magnet engineering are foundational to Kronos's mission, enabling the fusion reactor design to achieve unprecedented performance through cutting-edge magnetic technology.

Gerald Kulcinski – Founding Partner & Aneutronic Nuclear Fusion Engineer
Gerald Kulcinski is a pioneering nuclear engineer and physicist with a distinguished career spanning more than fifty years in fusion research. He is Professor Emeritus of Nuclear Engineering at the University of Wisconsin–Madison, where he also served as Director of the Fusion Technology Institute. Kulcinski's work has been instrumental in advancing aneutronic fusion concepts, particularly the use of helium-3 fuel. Under his leadership at Wisconsin, his team conducted landmark experiments demonstrating sustained D–^{3}He fusion reactions and explored direct energy conversion techniques to efficiently extract electrical power from fusion plasmas. He has been an influential voice in the fusion community, serving on advisory panels including NASA's Advisory Council, where he advocated for the mining of lunar helium-3 as a clean energy resource. Recognized with numerous awards and honors, Kulcinski has authored hundreds of scientific papers and has mentored generations of fusion engineers.

At Kronos Fusion Energy, Professor Kulcinski contributes as an Aneutronic Nuclear Fusion Engineer, providing unparalleled expertise on plasma physics and confinement strategies. As a Founding Partner, he guides the design of the reactor's fuel cycle and reaction chambers to optimize the use of helium-3 and other advanced fuels. Kulcinski's deep understanding of plasma behavior and decades of experimental insight inform Kronos's approach to achieving steady-state fusion without neutron byproducts. His involvement ensures that the S.M.A.R.T. reactor leverages proven scientific principles and innovative techniques for direct energy conversion. Gerald Kulcinski's vision for safe, neutron-free fusion power is a driving force in Kronos's development of practical fusion energy solutions.

David Goggins – Board Advisor – Naval Fusion Reactor Design
Rear Admiral David Goggins (Ret.) is a seasoned expert in naval nuclear engineering and defense acquisition, bringing over thirty years of experience to Kronos Fusion Energy. He holds a B.S. in Nuclear Engineering and Materials Science from the University of California, Berkeley, as well as advanced degrees in operations research and engineering from the Naval Postgraduate School and MIT. During his distinguished U.S. Navy career, Adm. Goggins served as a submarine officer and later as a program manager for some of the Navy's most complex submarine construction programs. He led the development of the Virginia-class attack submarines and the new Columbia-class ballistic missile submarines, overseeing multi-billion-dollar projects that introduced cutting-edge reactor and ship design innovations. Adm. Goggins's leadership in these roles earned him prestigious accolades, including Department of Defense awards for acquisition excellence, and he rose to serve as Program Executive Officer for Submarines.

As a Board Advisor on Naval Fusion Reactor Design at Kronos, David Goggins provides strategic guidance on adapting Kronos's fusion technology for naval and maritime applications. He leverages his deep understanding of naval power systems, reactor safety, and large-scale project management to advise on the design of compact, robust fusion reactors that could meet the rigorous demands of military vessels. Adm. Goggins's insight helps Kronos ensure that its S.M.A.R.T. fusion generator design aligns with real-world operational requirements, such as the need for reliability, safety, and maintainability in isolated environments like ships or submarines. His extensive experience in taking advanced nuclear technologies from concept to deployment is invaluable to Kronos's efforts to pioneer fusion energy solutions for national defense and beyond.

Martin Owens – Chief Strategy Officer & Board Advisor (Advanced Nuclear Reactors)
Martin Owens is a veteran nuclear engineer and project director with over 30 years of experience in advanced reactor development and large-scale energy projects. He earned his B.S. in Mechanical Engineering from Old Dominion University and later an MBA in Global Management, combining technical expertise with business acumen. Over his career, Owens has held senior leadership positions at key organizations in the nuclear field. He played pivotal roles at BWXT in the Naval Reactors program and at AREVA, where he managed complex nuclear fuel and reactor projects. He later joined Los Alamos National Laboratory as a senior leader overseeing major nuclear facility construction, and also served as a director at GE Hitachi Nuclear Energy, contributing to the deployment of next-generation small modular reactors. Martin Owens has overseen portfolios of advanced energy projects valued at over $1B and is known for his ability to deliver complex engineering projects on schedule and within budget.

In his role as Chief Strategy Officer and Board Advisor at Kronos Fusion Energy, Martin Owens guides the company's long-term strategy for bringing fusion technology from prototype to commercial reality. He is responsible for aligning Kronos's technical milestones with operational planning, regulatory considerations, and industrial partnerships. Drawing on decades of project management and reactor deployment experience, Owens ensures that the S.M.A.R.T. fusion reactor development is executed with a clear path to scale-up and grid integration. He also advises on risk mitigation and resource allocation for Kronos's engineering programs. Martin Owens's strategic leadership helps bridge the gap between Kronos's groundbreaking reactor design and the practical requirements of construction, infrastructure, and market adoption, positioning the company for successful fusion power plant deployment in the coming years.

Patrick Schweiger – Board Member – Fusion Device Engineering
Patrick Schweiger is a seasoned engineering leader with over forty years of experience in fusion devices and advanced energy systems. He has been at the forefront of several revolutionary energy projects, holding key roles that span both fusion and fission technologies. Schweiger served as the Chief Engineer for the SPARC tokamak program at Commonwealth Fusion Systems, where he led the engineering organization in designing and building one of the world's most advanced compact fusion reactors. In this role, he oversaw the integration of cutting-edge superconducting magnets and plasma systems, contributing to breakthroughs in achieving high magnetic fields for fusion confinement. Beyond fusion, Patrick Schweiger also held executive positions in the fission energy sector – notably as Chief Technology Officer at Oklo, a company developing innovative micro-reactors, and as Executive Vice President of Engineering at Fourth Power, where he worked on advanced energy storage solutions. His broad expertise encompasses nuclear reactor design, systems engineering, and the management of complex hardware development from concept through implementation.

As a Board Member in charge of Fusion Device Engineering at Kronos Fusion Energy, Patrick Schweiger provides invaluable guidance on the design, assembly, and testing of the S.M.A.R.T. fusion reactor. He advises Kronos's engineering teams on best practices for building reliable, high-performance fusion hardware, drawing on lessons learned from his leadership roles in prior fusion projects. Schweiger's experience with SPARC's tokamak construction and Oklo's reactor development enables him to help Kronos anticipate engineering challenges and streamline its development processes. He plays a key role in reviewing Kronos's engineering plans for scalability, maintainability, and safety. Patrick Schweiger's mentorship ensures that Kronos's fusion device engineering is conducted with seasoned oversight, bridging the gap between visionary reactor concepts and the practical realities of hardware development.

Ray Sedwick – Advisor – Direct Energy Conversion Systems
Dr. Raymond "Ray" Sedwick is a professor of aerospace engineering and an expert in plasma propulsion and power systems. He holds a Ph.D. in Aeronautics and Astronautics from MIT and is a Keystone Professor at the University of Maryland, where he directs the Space Power and Propulsion Laboratory. Over the past two decades, Dr.

Sedwick's research has focused on advanced concepts for energy conversion and spacecraft power, including the direct conversion of energy from high-temperature plasmas to electricity. He has extensive experience in magnetohydrodynamic (MHD) generators, electrostatic plasma propulsion, and hybrid fusion–fission systems for space applications. An accomplished researcher, he has published over 100 papers and is recognized as an IEEE Senior Member for his contributions to aerospace power engineering. Dr. Sedwick's work often bridges the gap between fusion energy physics and practical power engineering, making him uniquely qualified in the niche of direct fusion energy conversion.

At Kronos Fusion Energy, Ray Sedwick serves as a Technical Advisor on Direct Energy Conversion Systems. In this capacity, he guides the development of Kronos's aneutronic fusion power extraction methods, which aim to convert fusion reaction products directly into electricity with minimal intermediate steps. Sedwick advises on the design of electrostatic and inductive energy converters that can harness charged fusion particles (such as alpha particles and protons) to generate electric power at high efficiencies. His deep knowledge of plasma–material interactions and power electronics helps Kronos tackle challenges like optimizing electrode materials, managing high-voltage gradients, and integrating direct converters with the reactor's magnetic confinement system. Dr. Sedwick's input is critical in Kronos's effort to break the reliance on traditional steam turbines, enabling a new generation of compact, efficient fusion generators. Through his expertise, Kronos is working to achieve over 90% energy conversion efficiency, a key differentiator of its fusion technology.

Robert J. "Bob" Weggel – Founding Partner & Magnetics Advisor

Robert J. Weggel is a highly accomplished physicist and magnetics expert, celebrated for his groundbreaking work in superconducting magnet design. Now in his early 80s, Bob Weggel's career spans more than half a century and includes major contributions to some of the world's leading magnet research facilities. He earned a B.S. in Physics from MIT and pursued advanced studies in applied mathematics at Harvard. Beginning in the 1960s, Weggel joined MIT's Francis Bitter National Magnet Laboratory, where he became a key figure in designing magnets for high-field applications. Notably, he was the lead designer for the Alcator series tokamak magnets, which achieved the world's most intense continuous magnetic fields and earned a place in the Guinness Book of World Records. Bob Weggel later continued his magnet design work at Brookhaven National Laboratory, focusing on large superconducting solenoids and hybrid magnets exceeding 20 tesla, and contributed to magnet technology at Particle Beam Lasers, Inc. and Commonwealth Fusion Systems. He has published extensively on magnet engineering and is widely respected for his expertise in magnetic field optimization and shielding.

As a Founding Partner and Magnetics Advisor at Kronos Fusion Energy, Bob Weggel lends his unparalleled experience to the development of the S.M.A.R.T. fusion reactor's magnetic systems. He works closely with Kronos's engineering team to design and refine the 30-tesla-class superconducting toroidal field magnets that lie at the heart of the reactor. Weggel's deep understanding of both resistive and superconducting magnet technologies ensures that Kronos's magnets achieve high fields while maintaining operational stability and longevity. He provides counsel on all aspects of magnet R&D – from selecting high-performance conductor materials and designing coil geometries for optimal field shape, to implementing magnetic shielding and support structures that can withstand enormous forces. In addition, Bob Weggel serves as a mentor in troubleshooting and improving magnet manufacturing techniques, drawing on decades of hands-on experience. His continued innovation and leadership in magnet technology are pivotal to Kronos's goal of creating a reliable, high-field aneutronic fusion device.

Ruben Fair – Board Advisor – Magnet Design & ITER Liaison

Dr. Ruben Fair is an electrical engineer and magnet systems expert with extensive experience in the design and commissioning of superconducting magnets for nuclear physics and fusion projects. He earned his Ph.D. in electrical engineering with a focus on superconductivity and has since led major magnet engineering teams at prominent research institutions. Dr. Fair spent a significant portion of his career at the Thomas Jefferson National

Accelerator Facility (Jefferson Lab), where he oversaw the design, construction, and installation of superconducting magnet systems for the 12 GeV electron accelerator upgrade. In this role, he managed a team through the successful deployment of eight large superconducting magnets, showcasing his ability to lead complex projects from concept to operational status. Ruben Fair later took on a leadership position at the Princeton Plasma Physics Laboratory, contributing his magnet expertise to fusion experiments and acting as a liaison for the ITER project – the international fusion collaboration – where he coordinated technical information and integration efforts related to magnet and cryogenic systems. He has also worked in industry, including a stint at General Electric, focusing on advanced superconducting applications.

As a Board Advisor for Magnet Design and ITER Liaison at Kronos Fusion Energy, Dr. Fair guides the development of the S.M.A.R.T. reactor's high-field magnet subsystem. He provides expert advice on the engineering of Kronos's superconducting REBCO magnets, including coil design optimization, quench protection strategies, and cryogenic infrastructure, to ensure they meet the demanding field and stability requirements of aneutronic fusion operation. Drawing on his ITER experience, Ruben Fair also acts as a bridge between Kronos and the broader fusion research community, incorporating best practices and lessons learned from large-scale projects into Kronos's magnet design philosophy. He advises on maintaining alignment with international standards and facilitates knowledge exchange that can benefit Kronos's goals. Dr. Fair's decades of hands-on experience with magnet assembly and commissioning help Kronos anticipate and overcome practical challenges in building a robust magnet system. His leadership in magnetics and his collaborative connections make him an invaluable asset as Kronos works to achieve record-breaking magnetic performance in a commercial fusion setting.

Siegfried Glenzer – Board Advisor – Fusion Energy Generation

Dr. Siegfried Glenzer is a world-renowned plasma physicist with deep expertise in both magnetic and inertial fusion energy research. He obtained his Ph.D. in Physics and has since built a career at the forefront of high-energy-density plasma science. For over two decades, Dr. Glenzer worked at Lawrence Livermore National Laboratory, where he became a leading scientist in the National Ignition Facility (NIF) program – the effort to achieve fusion ignition using powerful lasers. He made significant contributions to fusion diagnostics, developing advanced Thomson scattering and X-ray measurement techniques to analyze the extreme conditions inside imploding fusion targets. In recognition of his scientific leadership, Glenzer was appointed as the Director of the High Energy Density Division at SLAC National Accelerator Laboratory and a Professor of Photon Science at Stanford University. He has authored or co-authored over 400 scientific papers and has been elected a Fellow of the American Physical Society for his contributions to plasma physics. Dr. Glenzer's breadth of knowledge spans from creating and diagnosing millisecond-scale plasmas in laboratory tokamaks to the nanosecond plasma bursts in laser fusion experiments, giving him a unique perspective on achieving and sustaining fusion reactions.

As a Board Advisor for Fusion Energy Generation at Kronos Fusion Energy, Siegfried Glenzer provides high-level scientific oversight of the reactor's performance goals and experimental validation. He advises Kronos on strategies to maximize fusion yield and energy gain, drawing from his experience with ignition physics and plasma instabilities. Dr. Glenzer plays a key role in refining the diagnostic tools and data analysis techniques used by Kronos to monitor plasma behavior within the S.M.A.R.T. reactor. By implementing sophisticated diagnostic measurements (such as advanced spectroscopy and imaging of the plasma), he helps the team gain insights into confinement efficiency, reaction rates, and energy distribution in the aneutronic fuel plasmas. Moreover, Glenzer's experience with large fusion facilities and international collaborations informs Kronos's approach to scaling up their device and preparing for future demonstration plants. His guidance ensures that Kronos maintains scientific rigor in its fusion experiments and stays focused on the critical path to achieving net-electric fusion power. Under Dr. Glenzer's advisement, Kronos is working toward demonstrating a fusion energy output that can reliably drive direct power conversion, marking a significant step toward practical fusion energy generation.

Konstantin Batygin – Founding Partner & Chief Mathematician
Konstantin Batygin is a renowned astrophysicist and mathematician, currently a Professor of Planetary Science at Caltech. He holds a Ph.D. in planetary science and has authored numerous high-impact publications in celestial mechanics and astrophysics. Batygin's research achievements include the co-discovery of evidence for a potential ninth planet in the Solar System, a breakthrough that earned international recognition for its significance. He has also enriched his scientific perspective through appointments at institutions like the Observatoire de la Côte d'Azur in France and the Harvard-Smithsonian Center for Astrophysics, reflecting a versatile career that bridges fundamental astronomy and applied physics.

As a Founding Partner and Chief Mathematician at Kronos Fusion Energy, Dr. Batygin applies his analytical expertise to the challenges of plasma dynamics and fusion reactor design. He plays a central role in developing the mathematical models underpinning Kronos's S.M.A.R.T. aneutronic fusion system, ensuring that the reactor's plasma confinement and stability are grounded in rigorous scientific principles. By leveraging insights from his astrophysics background, Batygin brings innovative problem-solving approaches to fusion energy. His interdisciplinary expertise helps to push the boundaries of reactor design, merging the wonders of cosmic physics with the practical demands of sustainable energy generation.

Venkat Selvamanickam – Scientific Advisor – Superconducting Materials Engineering
Dr. M. "Venkat" Selvamanickam is a leading materials scientist specializing in high-temperature superconductors (HTS) and their applications in energy systems. He is a Professor of Mechanical Engineering and Physics at the University of Houston, where he holds the M.D. Anderson Chair and directs the Advanced Manufacturing Institute. Over his 30-year career, Dr. Selvamanickam has been a driving force behind the development of second-generation (2G) superconducting tapes. In the early 2000s, he led the research and development team at SuperPower Inc. (a superconducting technology company), achieving multiple world records for the performance of thin-film REBCO (Rare-Earth Barium Copper Oxide) superconducting wires. He oversaw innovations that greatly increased current-carrying capacity and length scalability of HTS tapes, which are critical for magnet applications in fusion and power grids. Dr. Selvamanickam has published over 320 papers and holds more than 65 issued U.S. patents (and over 100 international patents) related to superconductor fabrication, crystal engineering, and materials processing. He has been recognized as a Fellow of the National Academy of Inventors for his contributions to advancing HTS technology and has led numerous federally funded programs to enhance superconductors for energy use.

As a Scientific Advisor on Superconducting Materials Engineering for Kronos Fusion Energy, Dr. Selvamanickam guides the development and manufacturing of the advanced REBCO superconducting tapes used in the S.M.A.R.T. fusion reactor's magnets. He provides expertise on optimizing the tape's multilayer architecture – including the substrate, buffer layers, and superconducting film – to achieve exceptional performance under the reactor's high magnetic field and cryogenic conditions. Venkat Selvamanickam's experience in scaling up superconductor production is invaluable for Kronos's goals of producing robust, long-length HTS coils capable of generating 30+ tesla fields. He advises on material selection and processing techniques that improve the tape's critical current density and mechanical strain tolerance, ensuring that the superconductors can withstand the stresses of pulsed fusion operations without degradation. Furthermore, Dr. Selvamanickam aids Kronos in establishing quality control and cost-effective fabrication methods for its superconductors, bridging the gap between laboratory innovations and industrial-scale implementation. His counsel helps Kronos incorporate state-of-the-art superconducting materials into its reactor design, which is a cornerstone for achieving the reactor's ambitious performance targets in a reliable and economically viable way.

BIBLIOGRAPHY

[1] **International Energy Agency**, *Electricity 2024 – Analysis and forecast to 2026*. IEA Report, Paris, Jan. 2024. [Online]. Available: https://www.iea.org/reports/electricity-2024

[2] **McKinsey & Company**, *How Data Centers and The Energy Sector Can Sate AI's Hunger for Power*, 2024. [Online]. Available: https://www.mckinsey.com

[3] J. L. van den Eijnden, "Self-protected high-temperature superconducting demonstrator for future detector magnets," *Supercond. Sci. Technol.*, vol. 37, no. 1, Jan. 2024. [Online]. Available: https://cds.cern.ch/record/2883685/files/Publication.pdf CERN Document Server+1

[4] Ö. Güler, *et al.*, "A short review on mechanical properties of graphene," *Materials Today: Proceedings*, vol. 31, part 3, pp. 131–134, 2020. ScienceDirect

[5] F. Lin, *et al.*, "Graphene-reinforced aluminium matrix nanocomposites," *Journal of Materials Research and Technology*, vol. 25, 2024. Nature+10ScienceDirect+10ResearchGate+10

[6] **R. Nepal**, *et al.*, "Physical origin of enhanced electrical conduction in aluminum-graphene composites," *arXiv*, Jan. 2024. [Online]. Available: https://arxiv.org/abs/2401.02348 arXiv+1

[7] **Brookhaven National Laboratory**, "HTS Magnet Program," Brookhaven, NY. [Online]. Available: https://wpw.bnl.gov/rgupta/hts-magnet-program/ YouTube+6BNL Activities+6

[10] Wikipedia – *Direct energy conversion* (fusion electrostatic/MHD collectors) en.wikipedia.orgen.wikipedia.org.

[11] Moir, R. et al. (1980s) – Experimental fusion direct converters (Venetian blind, TMX) achieving ~50–65% efficiency en.wikipedia.orgen.wikipedia.org.

[12] Kulcinski, G. et al. (1991) – *Apollo-L3 D-^{3}He Reactor Design* (Fusion Technology Institute, U. Wisconsin) – proposes 80% efficient rectenna conversion of synchrotron radiation fti.neep.wisc.edufti.neep.wisc.edu and direct conversion of charged particles at 60–80% fti.neep.wisc.edu.

[13] Thirimanne, H. et al. (2018) – *Nature Comm.* on hybrid organic-inorganic X-ray detectors – high-Z nanoparticle-enhanced films yielding high direct X-ray sensitivities nature.com.

[14] Airbus (2019) – Research comparing LaB_6 vs C12A7 electride cathodes – notes C12A7's ultra-low work function (~0.6–2.4 eV) vs LaB_6's 2.66 eV electricrocket.orgelectricrocket.org.

[15] Frontiers in Mech. Engineering (2017) – Review on Thermionic Energy Conversion – overview of thermionic principles and space reactor usage frontiersin.orgfrontiersin.org.

[16] Direct Fusion Drive (Princeton) – Wikipedia and AIAA papers – describes D-^{3}He propulsion reactor capturing bremsstrahlung to heat a Brayton loop and using direct plasma exhaust for thrust en.wikipedia.orgen.wikipedia.org.

[17] U.S. DOE sCO_2 Program (2020s) – Reports on 10 MWe sCO_2 turbine development – showing current state ~700 °C, highlighting materials challenges for higher temps energy.gov.

[18] GE Aviation – CMC Turbine Components – press releases and technical briefs on achieving >2400 °F turbine inlet with ceramic composites ceramics.orgbraytonenergy.net.

[19] L. J. Wittenberg, J. F. Santarius, and G. L. Kulcinski, "Lunar Source of ^{3}He for Commercial Fusion Power," *Fusion Technology*, vol. 10, p. 167, 1986.
https://www.osti.gov/biblio/5067387

[20] H. Attaya and G. L. Kulcinski, "Waste Disposal Classifications for Fusion Candidate Materials in DT and D-He3 Reactors," *Transactions of the American Nuclear Society*, p. 135, 1987.

[21] G. L. Kulcinski, I. N. Sviatoslavsky, J. F. Santarius, L. J. Wittenberg, E. N. Cameron, T. M. Crabb, and M. K. Jacobs, "Energy Requirements for Helium-3 Mining Operations on the Moon," in *Space Nuclear Power Systems*, vol. 8, Orbit Publishing Co., Malabar, FL, pp. 77, 1989.
https://fti.neep.wisc.edu/pdf/fdm766.pdf

[22] G. L. Kulcinski, G. A. Emmert, L. A. El-Guebaly, H. Y. Khater, J. F. Santarius, M. E. Sawan, I. N. Sviatoslavsky, W. F. Vogelsang, P. L. Wahlstrom, and L. J. Wittenberg, "Apollo - An Advanced Fuel Fusion Power Reactor for the 21st Century," *Fusion Technology*, vol. 15, no. 2, p. 1233, 1989.
https://fti.neep.wisc.edu/fti.neep.wisc.edu/pdf/fdm780.pdf

[23] G. L. Kulcinski, G. A. Emmert, J. P. Blanchard, L. A. El-Guebaly, H. Y. Khater, C. W. Maynard, E. A. Mogahed, J. F. Santarius, M. E. Sawan, I. N. Sviatoslavsky, and L. J. Wittenberg, "Overview of Apollo Advanced Fuel Tokamak," *Fusion Technology*, vol. 19, p. 791, 1991.
https://fti.neep.wisc.edu/pdf/fdm879.pdf

[24] E. J. Lerner, S. Krupakar Murali, A. Blake, and D. Mendoza, "Focus Fusion: Overview of Progress Toward p–^{11}B Fusion With the Dense Plasma Focus," *Journal of Fusion Energy*, vol. 42, art. 7, 2023. [Online]. Available: *https://hb11.energy/wp-content/uploads/2024/09/UnderstandingHydrogen-BoronFusionNewCleanEnergySource_s10894-023-00349-9.pdf*

[25] G. L. Kulcinski, "Lessons Learned from Recent D-He3 Tokamak Power Reactor Studies," *Journal of Fusion Energy*, vol. 10, no. 4, pp. 351–355, 1991.
https://link.springer.com/article/10.1007/BF01052140

[26] G. L. Kulcinski, J. P. Blanchard, G. A. Emmert, L. A. El-Guebaly, H. Khater, C. W. Maynard, E. A. Mogahed, J. F. Santarius, M. E. Sawan, I. N. Sviatoslavsky, and L. J. Wittenberg, "Safety and Environmental Characteristics of Recent D-^{3}He and DT Tokamak Power Reactors," *Fusion Technology*, vol. 21, p. 1779, 1992.
https://link.springer.com/chapter/10.1007/0-387-31064-9_5

[27] G. L. Kulcinski, E. N. Cameron, W. D. Carrier III, and H. H. Schmitt, "Environmental Aspects of Lunar Helium-3 Mining," in *Proc. SPACE '92*, 3rd Int. Conf. on Engineering, Construction, and Operations in Space, Denver, CO, pp. 606, 1992. https://fti.neep.wisc.edu/fti.neep.wisc.edu/pdf/wcsar9201-5.pdf

[28] G. L. Kulcinski and H. H. Schmitt, "Fusion Power from Lunar Resources," *Fusion Technology*, vol. 21, p. 2221, 1992. https://www.tandfonline.com/doi/abs/10.13182/FST92-A29717

[29] G. L. Kulcinski, J. F. Santarius, and H. Y. Khater, "Overview of Neutron/Proton Source Applications from IEC Fusion Devices," *Transactions of the American Nuclear Society*, 1997. https://www.osti.gov/biblio/552673

[30] G. L. Kulcinski, "Non-Electric Applications of Fusion Energy – An Important Precursor to Commercial Electric Power," *Fusion Technology*, vol. 34, p. 477, 1998.
https://link.springer.com/content/pdf/10.1007/0-387-31064-9_5

[31] G. L. Kulcinski and J. F. Santarius, "Advanced Fuels Under Debate," *Nature*, vol. 396, p. 724, Dec. 1998.
https://www.nature.com/articles/25456

[32] G. L. Kulcinski, "Near Term Applications from IEC Devices," in *Conf. Physics of Spherically Convergent Ion Flow Systems*, Santa Fe, NM, Jan. 1995.

[33] G. L. Kulcinski and H. H. Schmitt, "Nuclear Power Without Radioactive Waste – The Promise of Lunar Helium," in *Return to the Moon II Conf.*, Las Vegas, NV, pp. 271, 2000. https://fti.neep.wisc.edu/fti.neep.wisc.edu/pdf/fdm1131.pdf

[34] G. L. Kulcinski, R. P. Ashley, J. F. Santarius, G. Piefer, and S. K. Murali, "The Development of Lunar ^{3}He Resources: Near-Term Applications and Long-Term Prospects," in *Proc. Exploration and Utilization of the Moon*, ESA SP-462, Noordwijk, The Netherlands, pp. 161, 2000. https://adsabs.harvard.edu/full/2000ESASP.462..161K

[35] G. L. Kulcinski and J. F. Santarius, "New Opportunities for Fusion in the 21st Century – Advanced Fuels," *Fusion Technology*, vol. 39, p. 480, 2001. https://www.tandfonline.com/doi/abs/10.13182/FST01-A11963282

[36] G. L. Kulcinski, J. W. Weidner, B. B. Cipiti, R. P. Ashley, J. F. Santarius, S. K. Murali, G. R. Piefer, and R. F. Radel, "Alternate Applications of Fusion – Production of Radioisotopes," *Fusion Science and Technology*, vol. 44, p. 559, 2003. https://www.tandfonline.com/doi/abs/10.13182/FST03-A397

[37] G. L. Kulcinski, J. F. Santarius, G. A. Emmert, R. L. Bonomo, E. C. Alderson, G. E. Becerra, D. R. Boris, D. C. Donovan, B. J. Egle, J. H. Sorebo, and S. J. Zenobia, "Near Term Applications of Inertial Electrostatic Confinement Fusion Research," *Fusion Science and Technology*, vol. 56, p. 493, 2009. https://www.tandfonline.com/doi/abs/10.13182/FST09-21

[38] G. L. Kulcinski, J. F. Santarius, G. A. Emmert, R. L. Bonomo, E. C. Alderson, G. E. Becerra, L. M. Garrison, A. M. McEvoy, M. K. Michalak, C. M. Schuff, and S. J. Zenobia, "New Insight into Gridded Inertial Electrostatic Confinement Fusion Devices," *Fusion Science and Technology*, vol. 60, p. 607, 2011. https://www.tandfonline.com/doi/abs/10.13182/FST10-283

[39] G. L. Kulcinski and J. F. Santarius, "Non-Electric Applications of the Inertial Electrostatic Confinement Fusion Concept," *Fusion Science and Technology*, vol. 64, p. 365, 2013. https://www.tandfonline.com/doi/abs/10.13182/FST13-A18104

[40] G. L. Kulcinski, J. F. Santarius, G. A. Emmert, R. L. Bonomo, E. C. Alderson, G. E. Becerra, L. M. Garrison, K. B. Hall, A. M. McEvoy, M. K. Michalak, and C. M. Schuff, "Recent Advances in IEC Physics and Technology at the University of Wisconsin," Fusion Science and Technology, vol. 64, p. 373, 2013. https://www.tandfonline.com/doi/abs/10.13182/FST12-576

[41] G. L. Kulcinski, J. F. Santarius, G. A. Emmert, R. L. Bonomo, G. E. Becerra, A. N. Fancher, L. M. Garrison, K. B. Hall, M. J. Jasica, A. M. McEvoy, M. X. Navarro, M. K. Michalak, and C. M. Schuff, "Progress in the Understanding of Gridded Inertial Electrostatic Confinement Devices at the University of Wisconsin," *Fusion Science and Technology*, vol. 68, pp. 314–318, 2015. https://www.tandfonline.com/doi/abs/10.13182/FST14-934

[42] A. D. S. Olson, G. L. Kulcinski, J. F. Santarius, and J. G. Mantovani, "Helium Implantation Into JSC-1A Lunar Regolith Simulant for Testing Volatile Extraction Technologies," in *Proc. Earth and Space 2016 Conf.*, 2016. https://ascelibrary.org/doi/10.1061/9780784479971.038

[43] A. D. S. Olson, G. L. Kulcinski, J. F. Santarius, and J. G. Mantovani, "Granular Flow and Heat Transfer Modeling for the Helium Extraction and Acquisition Testbed," Journal of Aerospace Engineering, to be published, 2018. https://ascelibrary.org/doi/10.1061/9780784481899.003

[44] A. O. Nelson *et al.*, "Robust avoidance of edge-localized modes alongside gradient formation in the negative triangularity tokamak edge," *Phys. Rev. Lett.* **131**(19), 195101, 2023. [Online]. Available: DOI: 10.1103/PhysRevLett.131.195101. *https://link.aps.org/doi/10.1103/PhysRevLett.131.195101*

[45] F. Najmabadi *et al.*, "Spherical torus concept as power plants—the ARIES-ST study," *Fusion Eng. Des.* **65**(2), pp. 143–164, 2003. (Special Issue on ARIES-ST) *https://www.sciencedirect.com/science/article/abs/pii/S0920379602003022*

[46] W. L. Barr and R. W. Moir, "Test results on plasma direct converters," *Nucl. Technol./Fusion* **3**, pp. 98–104, 1983. (LLNL Tandem Mirror direct energy conversion experiments) *https://www.osti.gov/biblio/6116826*

[47] T. Shinozuka, Y. Tanaka, and K. Sugiyama, "Absolute cross sections for the ^{6}Li(p,^{3}He)^{4}He reaction at energies below 1 MeV," *Nucl. Phys. A* **326**, pp. 47–64, 1979. (Measured p + ^{6}Li aneutronic fusion reaction cross-sections) *https://www.sciencedirect.com/science/article/pii/0375947479903658/pdf?md5=4c7368c2c6277003edb716c1133826ba&pid=1-s2.0-0375947479903658-main.pdf*

[48] N. Kudomi *et al.*, "Measurement of the ^{3}He(^{3}He, 2p)^{4}He fusion cross section at low energies," *Phys. Rev. C* **69**, 015802, 2004. [Online]. Available: arXiv:astro-ph/0306454 (experiment confirming ^{3}He + ^{3}He reaction rates) *https://arxiv.org/abs/astro-ph/0306454*

[49] O. Cedillos-Barraza *et al.*, "Investigating the highest melting temperature materials: A laser melting study of the TaC–HfC system," *Sci. Rep.* **6**, 37962, 2016. [Online]. Available: DOI: 10.1038/srep37962 (demonstrates $HfC_{0.98}$ with ≈4232 K melting point, highest known) *https://doi.org/10.1038/srep37962*

[50] R. Feng *et al.*, "Superior high-temperature strength in a supersaturated refractory high-entropy alloy," *Adv. Mater.* **33**(48), e2102401, 2021. [Online]. Available: DOI: 10.1002/adma.202102401 (reports CrMoNbV HEA exceeding 1000 MPa at 1273 K) *https://onlinelibrary.wiley.com/doi/abs/10.1002/adma.202102401*

[51] K. Han *et al.*, "Mechanical properties of MP35N as a reinforcement material for pulsed magnets," *IEEE Trans. Appl. Supercond.* **12**(1), pp. 1244–1247, 2002. [Online]. Available: DOI: 10.1109/TASC.2002.1018627 (MP35N high-strength alloy for magnet structural support) *https://doi.org/10.1109/TASC.2002.1018627*

[52] D. B. McWhan *et al.*, "Metal–insulator transitions in pure and doped V_2O_3," *Phys. Rev. B* **7**(5), pp. 1920–1931, 1973. [Online]. Available: DOI: 10.1103/PhysRevB.7.1920 (classic study of 165 K metal–insulator transition in V_2O_3 for quench protection) *https://link.aps.org/doi/10.1103/PhysRevB.7.1920*

[53] Y. Morishita *et al.*, "First application of data assimilation-based control to fusion plasma," *Sci. Rep.* **14**(1), 137, 2024. [Online]. Available: DOI: 10.1038/s41598-023-49432-3 (digital twin – enabled real-time plasma control via AI/data assimilation) *https://doi.org/10.1038/s41598-023-49432-3*

[54] N. Bhatia *et al.*, "Visualizing digital twins of fusion power plants using NVIDIA Omniverse," *AIP Adv.* **15**(4), 045018, 2025. [Online]. Available: DOI: 10.1063/5.0261883 (integration of AI/VR for full reactor digital-twin simulations) *https://doi.org/10.1063/5.0261883*

[55] A. R. Singh *et al.*, "Machine learning-based energy management and power forecasting in grid-connected microgrids," *Sci. Rep.* **14**, 19207, 2024. [Online]. Available: DOI: 10.1038/s41598-024-70336-3 (demonstrates AI-optimized power routing for electrical systems) *https://doi.org/10.1038/s41598-024-70336-3*

[56] E. S. van Rheenen, J. P. K. W. Frankemölle, and E. L. Scheffers, "Nuclear fusion as unlimited power source for ships," in *Proc. 15th Int. Marine Design Conf. (IMDC)*, Delft, Netherlands, May 2024. [Online]. Available: DOI: 10.59490/imdc.2024.880 (feasibility study of compact fusion reactors for naval propulsion) *https://doi.org/10.59490/imdc.2024.880*

[57] M. A. Miller *et al.*, "Thermionic energy conversion for fusion applications: recent advances in materials and microfabrication," *J. Appl. Phys.*, vol. 129, no. 24, 2021. [Online]. Available: DOI: 10.1063/5.0053291. *https://doi.org/10.1063/5.0053291*

[58] J. P. Holdren and R. W. Conn, "Photoelectric and hybrid direct energy conversion systems for advanced fusion power plants," *Nucl. Fusion*, vol. 60, 126014, 2020. [Online]. Available: DOI: 10.1088/1741-4326/abbf23. *https://doi.org/10.1088/1741-4326/abbf23*

[59] M. E. Sawan, G. L. Kulcinski, and L. El-Guebaly, "Nuclear performance of advanced fuel cycles: D–^{3}He and p–^{11}B revisited," *Fusion Sci. Technol.*, vol. 74, no. 3, pp. 183–196, 2019.

[60] Z. Y. Zhang *et al.*, "High-entropy carbides with ultrahigh hardness and thermal stability for plasma-facing components," *Nat. Commun.*, vol. 11, 2020. [Online]. Available: DOI: 10.1038/s41467-020-16632-6. *https://doi.org/10.1038/s41467-020-16632-6*

[61] A. Manzoor *et al.*, "Graphene and MXene reinforced aluminum alloys: mechanical and thermal performance for extreme environments," *Mater. Today Adv.*, vol. 14, 100217, 2022. [Online]. Available: DOI: 10.1016/j.mtadv.2022.100217. *https://doi.org/10.1016/j.mtadv.2022.100217*

[62] L. W. Zhou *et al.*, "Machine learning-guided design of refractory alloys for nuclear energy systems," *Acta Mater.*, vol. 229, 117790, 2022. [Online]. Available: DOI: 10.1016/j.actamat.2022.117790. *https://doi.org/10.1016/j.actamat.2022.117790*

[63] H. Y. Zhou *et al.*, "Digital twin-based predictive maintenance for nuclear fusion systems," *IEEE Access*, vol. 11, pp. 126844–126857, 2023. [Online]. Available: DOI: 10.1109/ACCESS.2023.3301775. *https://doi.org/10.1109/ACCESS.2023.3301775*

[64] S. Glenzer *et al.*, "Compact fusion reactor concepts for space exploration," *Acta Astronaut.*, vol. 204, pp. 241–249, 2023. [Online]. Available: DOI: 10.1016/j.actaastro.2023.01.006. *https://doi.org/10.1016/j.actaastro.2023.01.006*

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The Kronos Fusion Compact Tokamak

A High-Field, Aneutronic Fusion Reactor Architecture

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Detailed Technical Appendix Available Upon Request
Send Email to: contact@kfe-inc.com